An Architecture for YANG-Push to Message Broker Integration

draft-ietf-nmop-yang-message-broker-integration

Motivation and architecture of a native
YANG-Push notifications and YANG Schema integration
into Message Broker and YANG Schema Registry

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IETF YANG-Push

A 22 years journey

2002

IAB Workshop

Defines operators' requirements in RFC 3535 to lifecycle CLI and SNMP. YANG, Netconf and Restconf development started.

2010

YANG 1.0

Specified in RFC 6020. 1.1 in RFC 7950.

2015

IETF YANG-Push Specification Started

Development of RFC 8639 and RFC 8641 started at IETF NETCONF.

2017

gNMI

gNMI was presented to IETF NETCONF and implementations started at major network vendors.

2019

IETF YANG-Push Specification Finished

Development of RFC 8639 and RFC 8641 concluded at IETF NETCONF without any major network vendor implementations.

2022

Data Mesh Integration

Vendor-specific implementations and IETF YANG-Push are hard to manage. New requirements emerged for integrating with the message broker and an automated data processing chain. New specifications are proposed to resolve these challenges.

2024

IETF YANG-Push Major Implementations Started

Questions arise. Proposing a simplified IETF YANG-Push and an Agile Incremental Driven Development.

Handling Operational YANG Modelled Data

State of the Union

Nowadays network operators are using **machine and human readable YANG** RFC 7950 to model their configurations and obtain YANG modelled data from their networks.

Network operators organize their data in a Data Mesh where a message broker such as Apache Kafka facilitates the exchange of messages among data processing components.

Today, subscribing to a YANG datastore, publishing a YANG modeled notifications message from the network and viewing the data in a time series database, **manual labor is needed to perform data transformation** to make a message broker and its data processing components with YANG notifications interoperable.

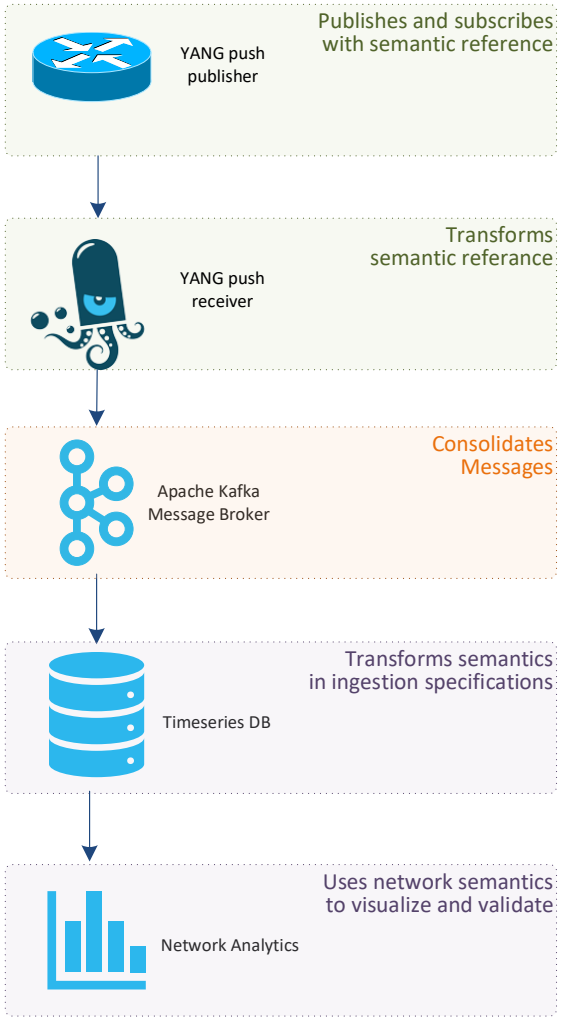
« Even though YANG is intented to ease data management, **this promise has not yet been fulfilled** for Network Telemetry RFC 9232 »

From YANG-Push to Network Analytics

Aiming for an automated data processing pipeline

- **A network operator aims for:**
  - An **automated data processing pipeline** which starts with YANG-Push, consolidates at Data Mesh and ends at Network Analytics.
  - Operational metrics where **IETF defines the semantics.**
  - Analytical metrics where **network operators gain actionable insights.**

- **We achieve this by integrating YANG-Push into Data Mesh to:**
  - Produce metrics from networks **with timestamps when network events were observed.**
  - Hostname, publisher ID and sequence numbers help us to understand **from where metrics were exported and measure its delay and loss.**
  - Forward **metrics unchanged** from networks
  - **Learn semantics** from networks and validate messages.
  - **Control semantic** changes end to end.



We have a dream

Digital Twin at your fingertips

« Imagine that your entire life as network engineer you have logged into routers to perform show commands to get a glimpse into the current state of your networks. »

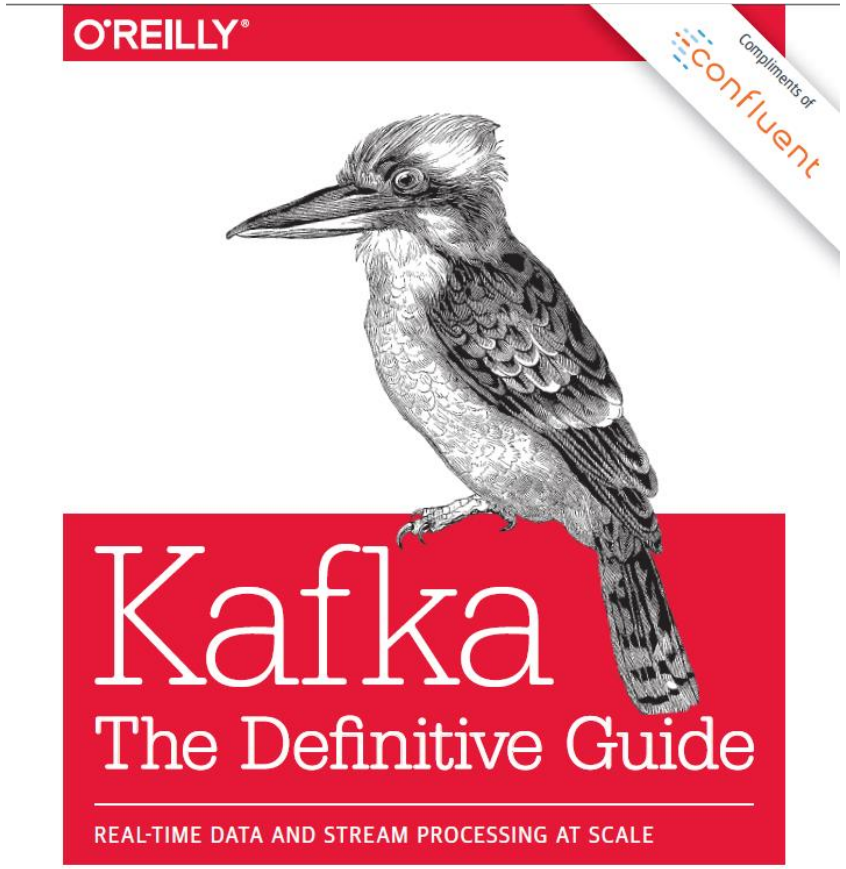
« Suddenly you see your colleague on the right querying the current network state in seconds directly from a real-time data stream. No access to routers needed. No databases needed. »



"NASA Mission Control Digital Twin in the 60s, Digital Twin is not Rocket Science"

Terminology (1)
Message Broker

- **Subject:** A named communication channel where a schema registry assigned schema id is associated.
- **Topic:** A communication channel for publishing and subscribing messages between producer and consumer with one or more subjects.
- **Topic Compaction:** The act of compressing messages in a topic to the latest state.
- **Partition:** Messages in a topic are spread over hash buckets where a hash bucket refers to a partition.
- **Message:** A piece of structured data sent between data processing components to facilitate communication in a distributed system
- **Message Key:** Metadata associated with a message to facilitate deterministic hash bucketing.
- **Segment:** A physical file containing multiple messages.



https://www.confluent.io/thank-you/resources/ebook/kafka-the-definitive-guide/

Terminology (2)

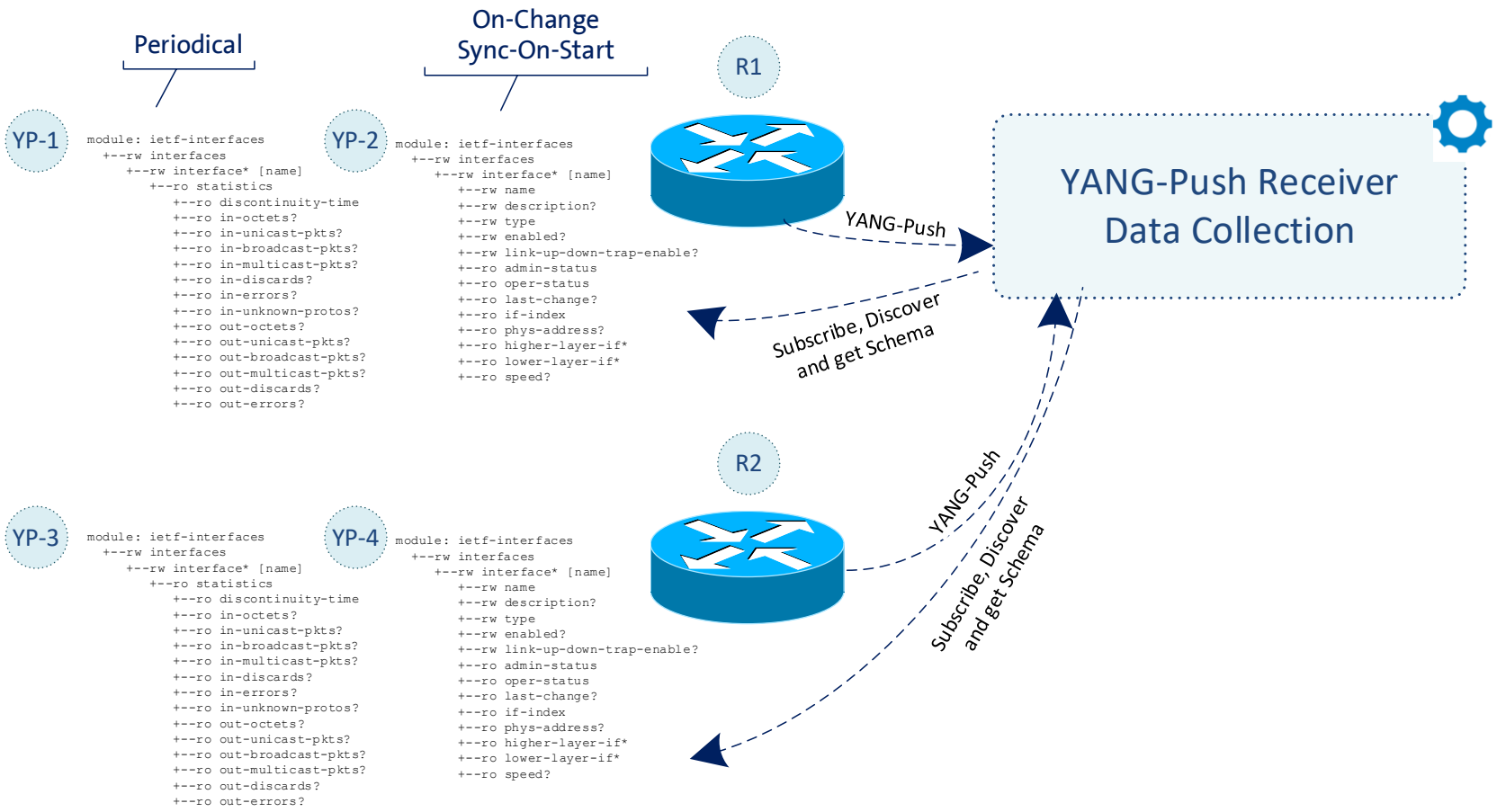
YANG Data Taxonomy and Index

- **Dimensional Data:** Structured information in a data store [Ralph Kimball]. Example: YANG with data taxonomy.
- **Data Taxonomy:** Is a hierarchical system for classifying and organizing data into categories and subcategories
- **YANG Schema Tree:** The definition hierarchy specified within a module.
- **YANG Schema Node:** A node in the schema tree.
- **YANG Item Identifiers:** Section 3.3 in RFC 9254 defines unique text-based identifiers for each YANG element including YANG schema nodes.
- **YANG index:** Is a subset of YANG item identifiers containing only YANG schema node identifiers.

```
module: ietf-interfaces
     +--rw interfaces
        +--rw interface* [name]
           +--rw name                            string
           +--rw description?                    string
           +--rw type                            identityref
           +--rw enabled?                        boolean
           +--rw link-up-down-trap-enable?       enumeration {if-mib}?
           +--ro admin-status                    enumeration {if-mib}?
           +--ro oper-status                     enumeration
           +--ro last-change?                    yang:date-and-time
           +--ro if-index                        int32 {if-mib}?
           +--ro phys-address?                   yang:phys-address
           +--ro higher-layer-if*                interface-ref
           +--ro lower-layer-if*                 interface-ref
           +--ro speed?                          yang:gauge64
           +--ro statistics
              +--ro discontinuity-time    yang:date-and-time
              +--ro in-octets?            yang:counter64
              +--ro in-unicast-pkts?      yang:counter64
              +--ro in-broadcast-pkts?    yang:counter64
              +--ro in-multicast-pkts?    yang:counter64
              +--ro in-discards?          yang:counter32
              +--ro in-errors?            yang:counter32
              +--ro in-unknown-protos?    yang:counter32
              +--ro out-octets?           yang:counter64
              +--ro out-unicast-pkts?     yang:counter64
              +--ro out-broadcast-pkts?   yang:counter64
              +--ro out-multicast-pkts?   yang:counter64
              +--ro out-discards?         yang:counter32
              +--ro out-errors?           yang:counter32
```

YANG-Push

Discover and Subscribe to YANG metrics



From discovering YANG-Push subscription capabilities defined in RFC 9196, subscribing interesting metrics periodical (**statistics**), on-change (**state changes**) or on-change with sync-on-start (**states**) defined in RFC 8641.

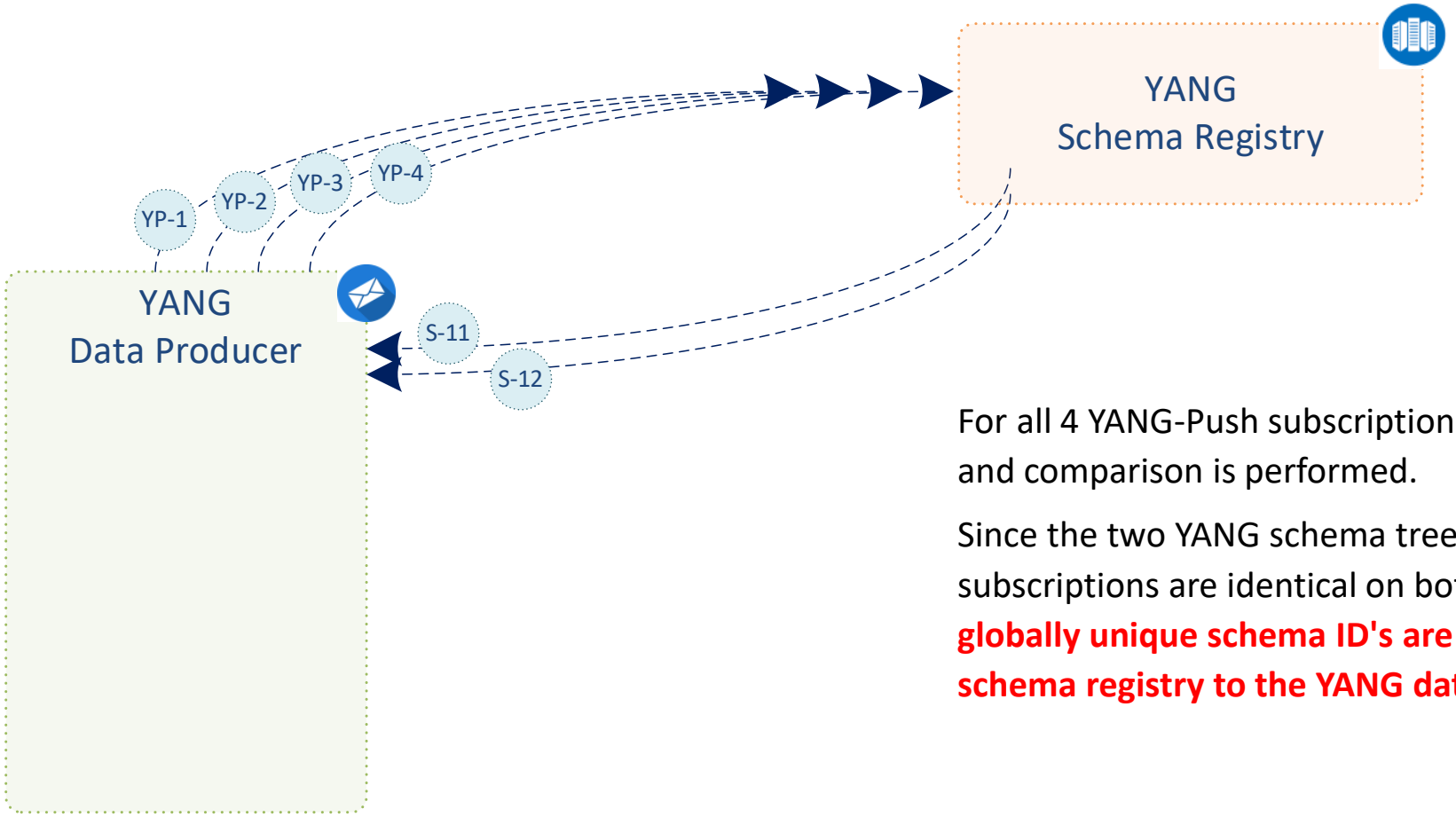
Each subscription refers to network node, datastore (RFC 8342) and a schema tree.

In this RFC 8343 example ietf-interface statistics are subscribed periodically and ietf-interface states on-change sync-on-start. **YANG-Push subscription ID's are per network node significant.**

Data Collection obtains for each subscription the YANG schema tree by leveraging <get-schema> (RFC 6022), YANG Library (RFC 8525) and draft-ietf-netconf-yang-library-augmentedby.

YANG Schema Registry

From 4 subscription ID's to 2 schema ID's

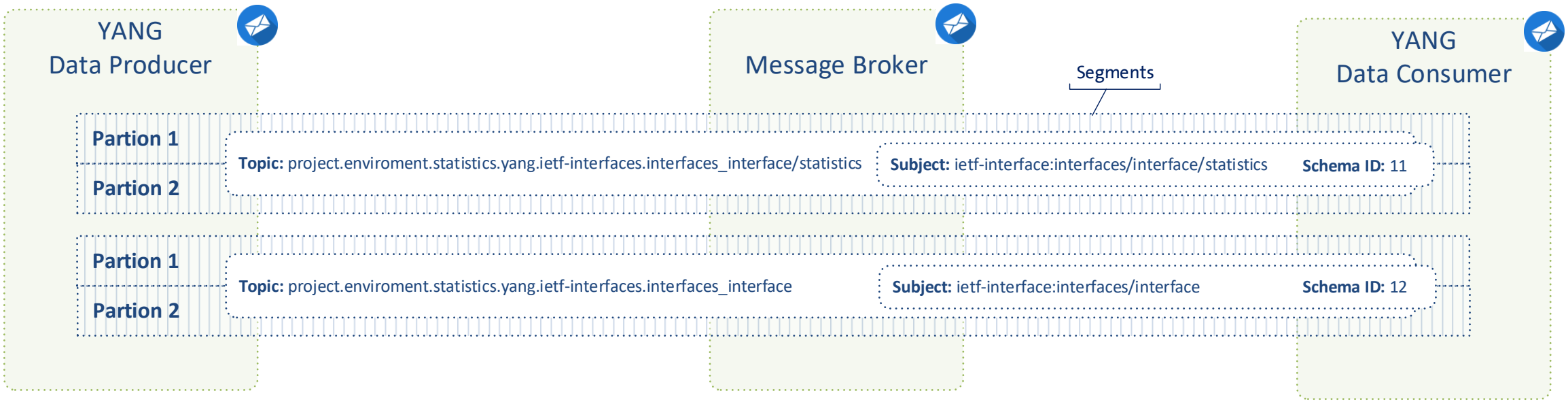


For all 4 YANG-Push subscriptions, YANG schema registration and comparison is performed.

Since the two YANG schema trees for both YANG-Push subscriptions are identical on both network nodes, **two YANG globally unique schema ID's are being issued from the YANG schema registry to the YANG data producer.**

Message Broker

Topics, Subjects, Partitions, Segments and Message Keys



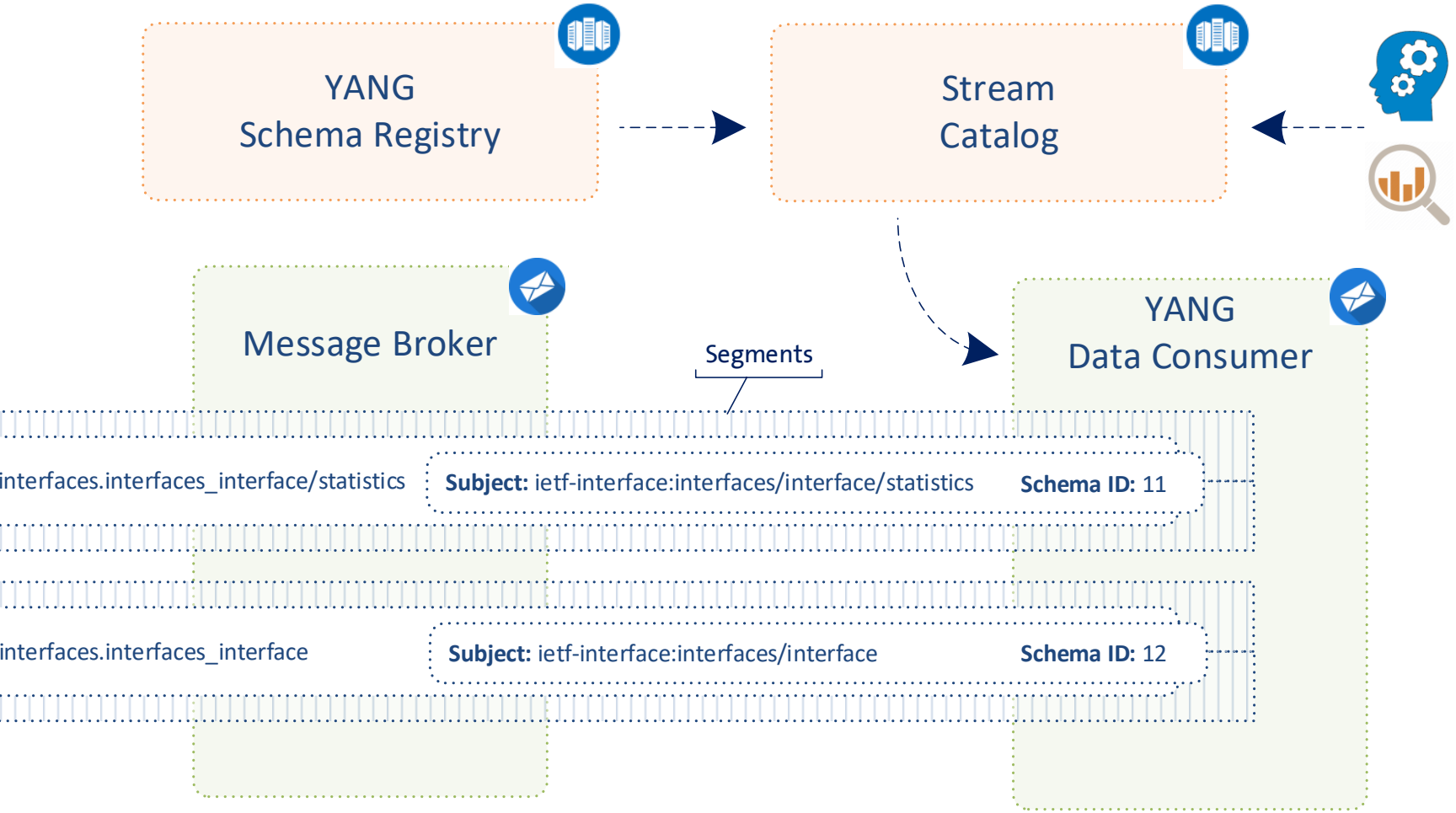
The YANG data producer creates for each YANG schema a new message broker topic, a message key and defines the number of partitions being used for the topic.

It serializes the message with the previously generated message key and message content according to draft-ietf-nmop-message-broker-telemetry-message.

Each message is prefixed with the previously obtained schema ID representing a unique message subject. The messages are distributed according to the hashed message key across the partitions into continuous segments.

YANG Data Consumption

Discover and Subscribe to YANG metrics



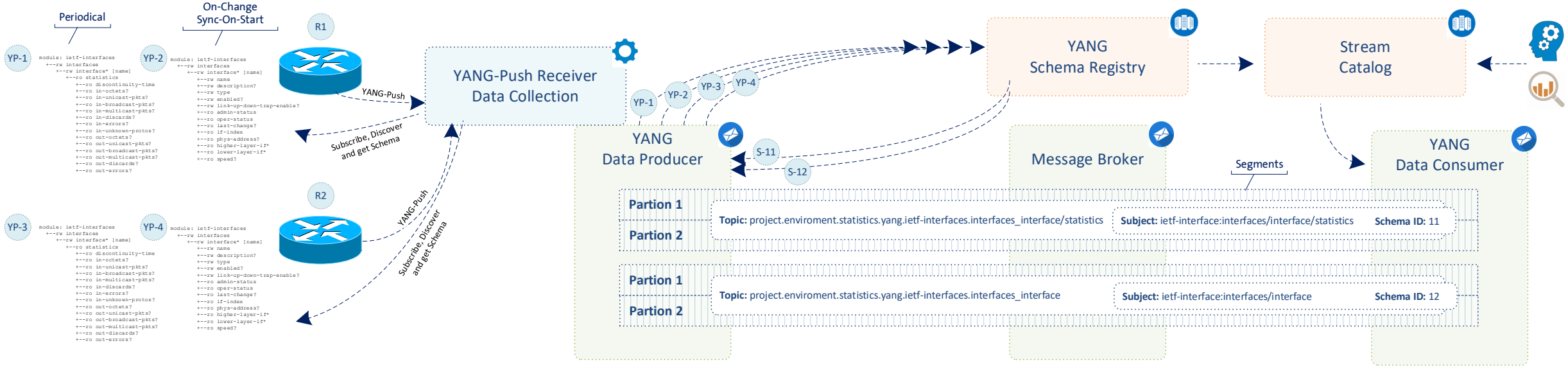
A user or AI application/agent subscribes discovers through the stream catalog interesting metrics and subscribes to message broker topic.

More than one topic can be consumed at once by using a wildcard such as: project.enviroment.states.yang.* to consume all YANG state metrics.

The consumer hashes the message key and applies modulo with the number of partitions to determine the partition it needs to consume from to obtain messages with desired message key.

Fully YANG aware Data Pipeline

From network nodes to message broker nodes



At the YANG data producer, thanks to the YANG awareness to the YANG data producer and the YANG schema registry, **for each unique YANG schema tree a message broker topic and subject is being created, and messages are being indexed and hashed according to the message key**.

At the message broker, thanks to the dedicated topic per YANG schema and its message key, **YANG state metrics can be compacted to the latest state; reducing the number of stored messages.**

At the YANG data consumer, thanks to the YANG awareness of the stream catalog and consumer, a specific topic, partition and subject can be consumed for the interested YANG metrics; **reducing the number of messages to be consumed.**

Addressing YANG Specification and Integration Gaps

12 documents at NMOP, NETCONF and NETMOD

YANG-Push Transport Gaps:

IESG • UDP-based Transport for Configured Subscriptions
draft-ietf-netconf-udp-notif

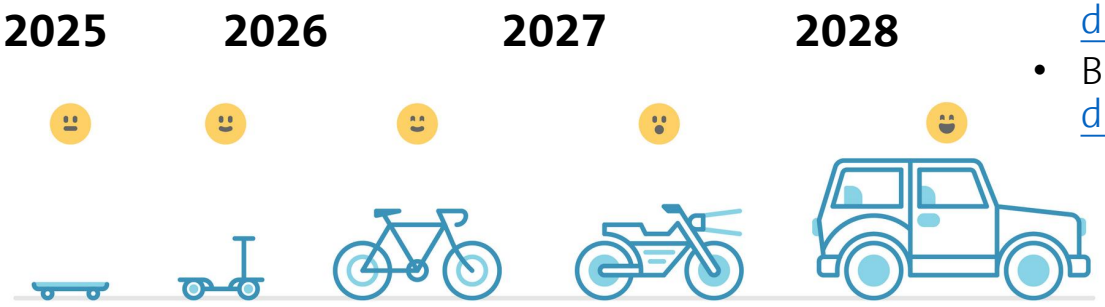
IESG • Subscription to Distributed Notifications
draft-ietf-netconf-distributed-notif

YANG-Push Specifications Gaps:

IESG • YANG Notification Transport Capabilities
draft-ietf-netconf-yp-transport-capabilities

IESG • Extensible YANG model for YANG-Push Notifications
draft-ietf-netconf-notif-envelope

• Validating anydata in YANG Library context
draft-ietf-netmod-yang-anydata-validation



YANG-Push Integration Gaps and Arch:

IESG • Support of Versioning in YANG Notifications Subscription
draft-ietf-netconf-yang-notifications-versioning

IESG • Augmented-by Addition into the IETF-YANG-Library
draft-ietf-netconf-yang-library-augmentation

YANG-Push Simplification:

• YANG-Push Operational Data Observability Enhancements
draft-ietf-netconf-yp-observability

YANG-Push Message Broker:

• An Architecture for YANG-Push to Message Broker Integration
draft-ietf-nmop-yang-message-broker-integration

• Extensible YANG Model for Network Telemetry Notifications
draft-ietf-nmop-message-broker-telemetry-message

• YANG Message Keys for Message Broker Integration
draft-ietf-nmop-yang-message-broker-message-key

• BMP YANG Model for Network Telemetry Messages
draft-netana-nmop-message-broker-bmp-telemetry-msg

IETF YANG-Push Implementations

Incremental development…

MVP 1 - Works

4 major vendor implementations finalized.

- draft-ietf-netconf-udp-notif transport (with segmentation option) and notifications encoded in RFC 7951 IETF-JSON.
- RFC 8641 periodic subscription with anchor-time configurable in ietf-subscribed-notifications.
- YANG notifications as defined in draft-netana-netconf-notif-envelope, YANG module name, version and yang-library-content-id as in draft-ietf-netconf-yang-notifications-versioning and notification capabilities (RFC 9196) discoverable as defined in their documents.
- YANG-Library as defined in RFC 8525 and draft-ietf-netconf-yang-library-augmentedby.

MVP 2 – Scales and Secures

2 major vendor implementations started.

- Distributed notifications as defined in draft-ietf-netconf-distributed-notif.
- YANG notifications encoded in CBOR (named identifiers) as defined in RFC 9254.
- YANG notifications DTLS (1.2 MUST RFC 6347, 1.3 SHOULD RFC 9147) encrypted as defined in draft-ietf-netconf-udp-notif.

MVP 3 - Optimizes

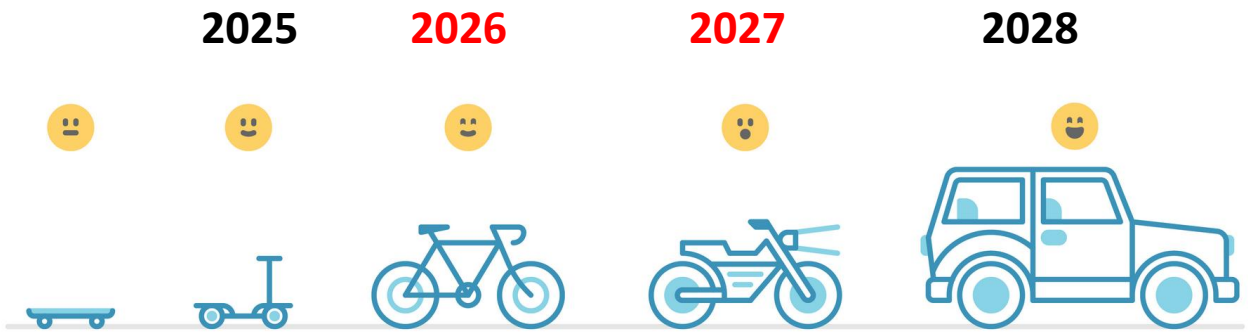
- RFC 8641 on-change subscription configurable in ietf-subscribed-notifications.
- RFC 8641 on-change and periodical subscription in RFC 9196 defined capabilities discoverable.
- draft-ietf-netconf-udp-notif transport, encoding and encryption capabilities (RFC 9196) discoverable with draft-ietf-netconf-yp-transport-capabilities.

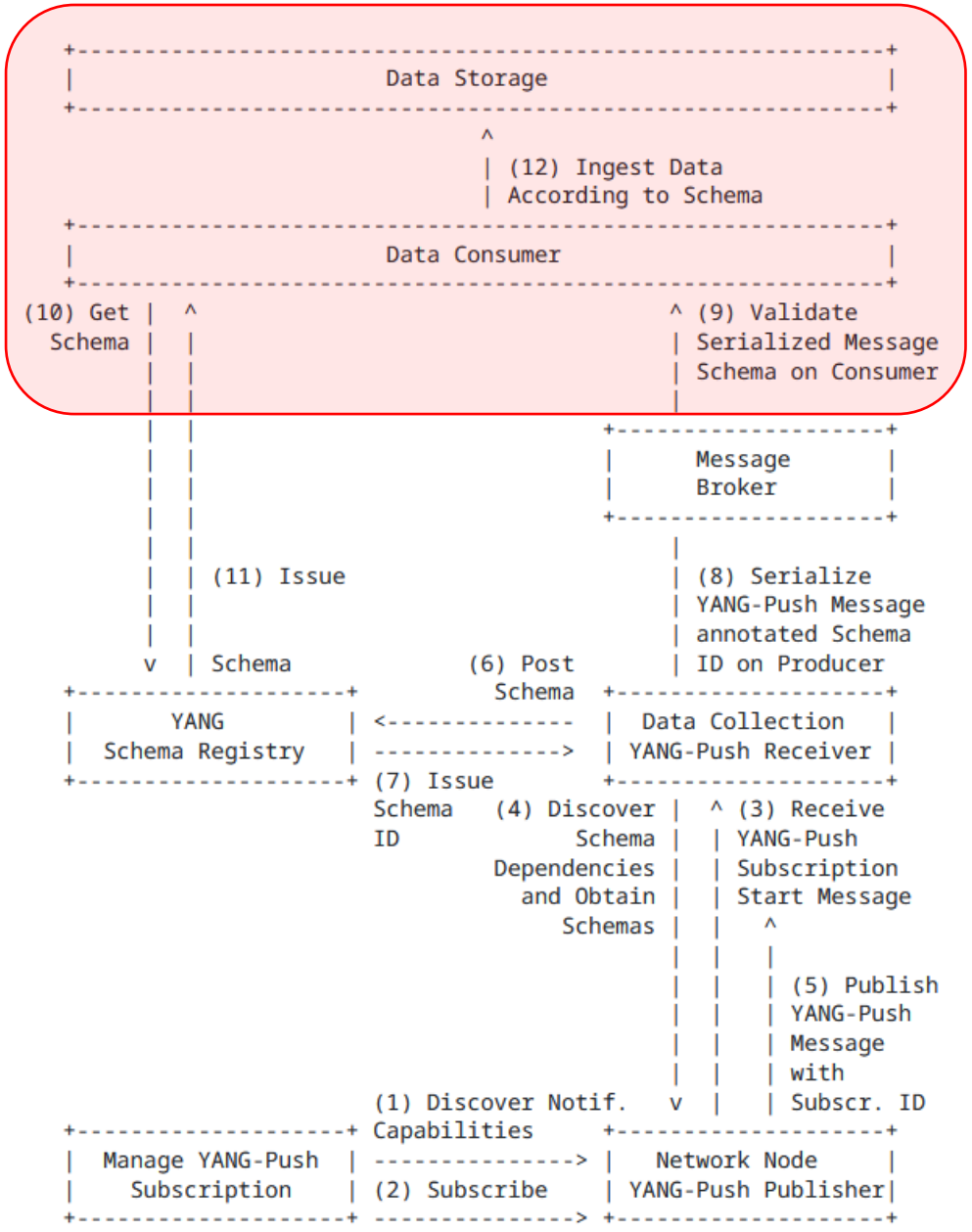
Part 2

Our collaboration goes into the next step in 2026

This allows Swisscom

to transition from SNMP to YANG





IETF YANG Model Implementations

Incremental development…

"Starting 2026, Swisscom onboards new network platforms with YANG-Push only. Initiating the SNMP to YANG lifecycle and the standardization of YANG models."

MVP 2 – Physical & Essential

Already supported by OLT MA5800.

- ietf-interfaces.yang (inventory, state and statistic)
- ietf-hardware.yang (inventory, state and statistic)
- ietf-alarms.yang (state)
- ieee802-dot1ab-lldp.yang (inventory and state)

MVP 3 – Logical Relationships

- ieee802-dot1ax.yang (inventory, state and statistic)
- ietf-bfd-ip-sh.yang (state)
- ietf-bfd-ip-mh.yang (state)
- ietf-bfd-lag.yang (state)
- ietf-isis.yang (state)



Gabriele Di Piazza and Markus Reber signing the Network Analytics MoU at MWC 2025

Blue Planet and Swisscom signed Network Analytics MoU

Thomas Graf
Distinguished Network Engineer and Network Analytics Architect at Swisscom

March 7, 2025

Dear colleagues,

I like to share a highlight of the 2025 Mobile World Congress.

It is a great honor and pleasure to announce that Blue Planet and Swisscom signed a Memorandum of Understanding in Network Analytics, collaborating in the area of:

• YANG-Push Messaging Integration into Apache Kafka Message Broker

Both companies understand the importance of moving network management lifecycle from using SNMP to using YANG for seamless integration into today's Data Mesh Big Data architectures.

RFC 8343 – ietf-interfaces.yang

YANG Data Model for Interface Management

```
module: ietf-interfaces
  +--rw interfaces
     +--rw interface* [name]
        +--rw name                            string
        +--rw description?                    string
        +--rw type                            identityref
        +--rw enabled?                        boolean
        +--rw link-up-down-trap-enable?       enumeration {if-mib}?
        +--ro admin-status                    enumeration {if-mib}?
        +--ro oper-status                     enumeration
        +--ro last-change?                    yang:date-and-time
        +--ro if-index                        int32 {if-mib}?
        +--ro phys-address?                   yang:phys-address
        +--ro higher-layer-if*                interface-ref
        +--ro lower-layer-if*                 interface-ref
        +--ro speed?                          yang:gauge64
        +--ro statistics
           +--ro discontinuity-time     yang:date-and-time
           +--ro in-octets?             yang:counter64
           +--ro in-unicast-pkts?       yang:counter64
           +--ro in-broadcast-pkts?     yang:counter64
           +--ro in-multicast-pkts?     yang:counter64
           +--ro in-discards?           yang:counter32
           +--ro in-errors?             yang:counter32
           +--ro in-unknown-protos?     yang:counter32
           +--ro out-octets?            yang:counter64
           +--ro out-unicast-pkts?      yang:counter64
           +--ro out-broadcast-pkts?    yang:counter64
           +--ro out-multicast-pkts?    yang:counter64
           +--ro out-discards?          yang:counter32
           +--ro out-errors?            yang:counter32
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

- For inventory and alert use cases the state of the interface.

and periodically

- For performance measurement use cases the interface statistics.

RFC 8348 – ietf-hardware.yang

YANG Data Model for Hardware Management

```
module: ietf-hardware
  +--rw hardware
     +--ro last-change?   yang:date-and-time
     +--rw component* [name]
        +--rw name                string
        +--rw class               identityref
        +--ro physical-index?     int32 {entity-mib}?
        +--ro description?        string
        +--rw parent?             -> ../../component/name
        +--rw parent-rel-pos?     int32
        +--ro contains-child*     -> ../../component/name
        +--ro hardware-rev?       string
        +--ro firmware-rev?       string
        +--ro software-rev?       string
        +--ro serial-num?         string
        +--ro mfg-name?           string
        +--ro model-name?         string
        +--rw alias?              string
        +--rw asset-id?           string
        +--ro is-fru?             boolean
        +--ro mfg-date?           yang:date-and-time
        +--rw uri*                inet:uri
        +--ro uuid?               yang:uuid
        +--rw state {hardware-state}?
        |  +--ro state-last-changed?   yang:date-and-time
        |  +--rw admin-state?          admin-state
        |  +--ro oper-state?           oper-state
        |  +--ro usage-state?          usage-state
        |  +--ro alarm-state?          alarm-state
        |  +--ro standby-state?        standby-state
        +--ro sensor-data {hardware-sensor}?
           +--ro value?                 sensor-value
           +--ro value-type?            sensor-value-type
           +--ro value-scale?           sensor-value-scale
           +--ro value-precision?       sensor-value-precision
           +--ro oper-status?           sensor-status
           +--ro units-display?         string
           +--ro value-timestamp?       yang:date-and-time
           +--ro value-update-rate?     uint32
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

- For inventory use cases the state of the hardware.
- For alert use cases the hardware state change notifications.

and periodically

- For performance measurement use cases the hardware sensor-data.

```
  notifications:
    +---n hardware-state-change
    +---n hardware-state-oper-enabled {hardware-state}?
    |  +--ro name?          -> /hardware/component/name
    |  +--ro admin-state?   -> /hardware/component/state/admin-state
    |  +--ro alarm-state?   -> /hardware/component/state/alarm-state
    +---n hardware-state-oper-disabled {hardware-state}?
       +--ro name?          -> /hardware/component/name
       +--ro admin-state?   -> /hardware/component/state/admin-state
       +--ro alarm-state?   -> /hardware/component/state/alarm-state
```

RFC 8632 – ietf-alarms.yang

YANG Data Model for Alarm Management

```
notifications:
    +---n alarm-notification
    |  +--ro resource                  resource
    |  +--ro alarm-type-id             alarm-type-id
    |  +--ro alarm-type-qualifier?     alarm-type-qualifier
    |  +--ro alt-resource*             resource
    |  +--ro related-alarm* [resource alarm-type-id alarm-type-
qualifier] {alarm-correlation}?
    |  |  +--ro resource                  -> /alarms/alarm-
list/alarm/resource
    |  |  +--ro alarm-type-id             -> /alarms/alarm-
list/alarm[resource=current()/../resource]/alarm-type-id
    |  |  +--ro alarm-type-qualifier      -> /alarms/alarm-
list/alarm[resource=current()/../resource][alarm-type-
id=current()/../alarm-type-id]/alarm-type-qualifier
    |  +--ro impacted-resource*        resource {service-impact-
analysis}?
    |  +--ro root-cause-resource*      resource {root-cause-analysis}?
    |  +--ro time                      yang:date-and-time
    |  +--ro perceived-severity        severity-with-clear
    |  +--ro alarm-text                alarm-text
    +---n alarm-inventory-changed
```

With IETF YANG-Push we like to subscribe

with on-change

- For alert use cases the alarm notifications.

IEEE LLDP – ieee802-dot1ab-lldp.yang

YANG Data Model for LLDP Neighbor Discovery

```
module: ieee802-dot1ab-lldp
  +--rw lldp
     +--ro local-system-data
     |  +--ro chassis-id-subtype?                 ieee:chassis-id-subtype-type
     |  +--ro chassis-id?                         ieee:chassis-id-type
     |  +--ro system-name?                        string
     |  +--ro system-description?                 string
     |  +--ro system-capabilities-supported?      lldp-types:system-capabilities-map
     |  +--ro system-capabilities-enabled?        lldp-types:system-capabilities-map
     +--rw port* [name dest-mac-address]
        +--rw name                              if:interface-ref
        +--rw dest-mac-address                  ieee:mac-address
        +--ro port-id?                          ieee:port-id-type
        +--ro remote-systems-data* [time-mark remote-index]
           +--ro time-mark                            yang:timeticks
           +--ro remote-index                         uint32
           +--ro remote-too-many-neighbors?           boolean
           +--ro remote-changes?                      boolean
           +--ro chassis-id-subtype?                  ieee:chassis-id-subtype-type
           +--ro chassis-id?                          ieee:chassis-id-type
           +--ro port-id-subtype?                     ieee:port-id-subtype-type
           +--ro port-id?                             ieee:port-id-type
           +--ro port-desc?                           string
           +--ro system-name?                         string
           +--ro system-description?                  string
           +--ro system-capabilities-supported?       lldp-types:system-capabilities-map
           +--ro system-capabilities-enabled?         lldp-types:system-capabilities-map
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

- For inventory use cases the local and remote-systems-data.

IEEE dot1ax – ieee802-dot1ax.yang

YANG Data Model for LAG Interface Management

```
module: ieee802-dot1ax
  +--rw lag-system
     +--rw aggregating-system* [agg-system]
        +--rw agg-system          string
        +--rw system-id?          ieee:mac-address
        +--rw system-priority?    uint32

  augment /if:interfaces/if:interface:
    +--rw aggregator
       +--rw name?                           string
       +--rw agg-system-name?                string
       +--rw admin-state?                    enumeration
       +--rw link-up-down-notification?      enumeration
       +--rw collector-max-delay?            int16
       +--rw aggregator-lacp
          +--rw actor-admin-key?    int16

  augment /if:interfaces/if:interface:
    +--rw aggregation-port
       +--rw aggregation-port-lacp
          +--rw actor-system-priority?           int16
          +--rw actor-admin-key?                 int16
          +--rw partner-admin-system-priority?   int16
          +--rw partner-admin-system-id?         ieee:mac-address
          +--rw partner-admin-key?               int16
          +--rw actor-port-priority?             int16
          +--rw partner-admin-port?              int16
          +--rw partner-admin-port-priority?     int16
          +--rw actor-admin-state?               bits
          +--rw partner-admin-state?             bits
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

- For inventory use cases the state of the hardware.

RFC 9127 – ietf-bfd.yang

YANG Data Model for Bidirectional Forwarding Detection

BFD IP Single-Hop Hierarchy

https://datatracker.ietf.org/doc/html/rfc9127#section-2.6

```
  notifications:
    +---n singlehop-notification
       +--ro local-discr?                   discriminator
       +--ro remote-discr?                  discriminator
       +--ro new-state?                     state
       +--ro state-change-reason?           iana-bfd-types:diagnostic
       +--ro time-of-last-state-change?     yang:date-and-time
       +--ro dest-addr?                     inet:ip-address
       +--ro source-addr?                   inet:ip-address
       +--ro session-index?                 uint32
       +--ro path-type?                     identityref
       +--ro interface?                     if:interface-ref
       +--ro echo-enabled?                  boolean
```

BFD-over-LAG Hierarchy

https://datatracker.ietf.org/doc/html/rfc9127#section-2

```
  notifications:
    +---n lag-notification
       +--ro local-discr?                   discriminator
       +--ro remote-discr?                  discriminator
       +--ro new-state?                     state
       +--ro state-change-reason?           iana-bfd-types:diagnostic
       +--ro time-of-last-state-change?     yang:date-and-time
       +--ro dest-addr?                     inet:ip-address
       +--ro source-addr?                   inet:ip-address
       +--ro session-index?                 uint32
       +--ro path-type?                     identityref
       +--ro lag-name?                      if:interface-ref
       +--ro member-link?                   if:interface-ref
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

- For alert use cases the hardware state notifications.

BFD IP Multihop Hierarchy

https://datatracker.ietf.org/doc/html/rfc9127#section-2.7

```
  notifications:
    +---n multihop-notification
       +--ro local-discr?                   discriminator
       +--ro remote-discr?                  discriminator
       +--ro new-state?                     state
       +--ro state-change-reason?           iana-bfd-types:diagnostic
       +--ro time-of-last-state-change?     yang:date-and-time
       +--ro dest-addr?                     inet:ip-address
       +--ro source-addr?                   inet:ip-address
       +--ro session-index?                 uint32
       +--ro path-type?                     identityref
```

RFC 9130 – ietf-isis.yang

YANG Data Model for the IS-IS Protocol

```
notifications:
  +---n database-overload
     |  +--ro routing-protocol-name?   -> /rt:routing/
     |  |                                 control-plane-protocols/
     |  |                                 control-plane-protocol/name
     |  +--ro isis-level?              level
     |  +--ro overload?                enumeration
     +---n if-state-change
     |  +--ro routing-protocol-name?   -> /rt:routing/
     |  |                                 control-plane-protocols/
     |  |                                 control-plane-protocol/name
     |  +--ro isis-level?              level
     |  +--ro interface-name?          if:interface-ref
     |  +--ro interface-level?         level
     |  +--ro extended-circuit-id?     extended-circuit-id
     |  +--ro state?                   if-state-type
     +---n adjacency-state-change
     |  +--ro routing-protocol-name?   -> /rt:routing/
     |  |                                 control-plane-protocols/
     |  |                                 control-plane-protocol/name
     |  +--ro isis-level?              level
     |  +--ro interface-name?          if:interface-ref
     |  +--ro interface-level?         level
     |  +--ro extended-circuit-id?     extended-circuit-id
     |  +--ro neighbor?                string
     |  +--ro neighbor-system-id?      system-id
     |  +--ro state?                   adj-state-type
     |  +--ro reason?                  string
```

With IETF YANG-Push we like to subscribe

with on-change sync-on-start

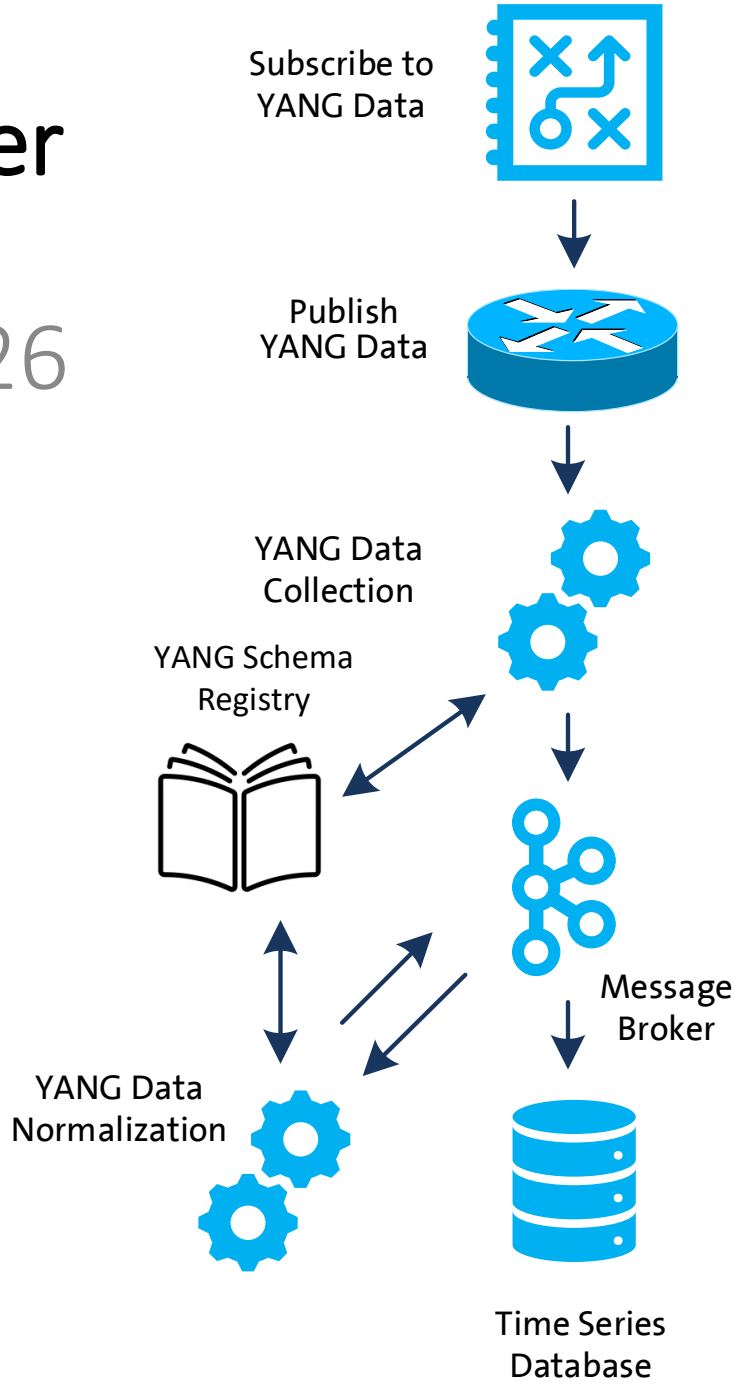
- For alert use cases the hardware state notifications.


Yang
Kafka

Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 126 Hackathon, July 18/19th 2026





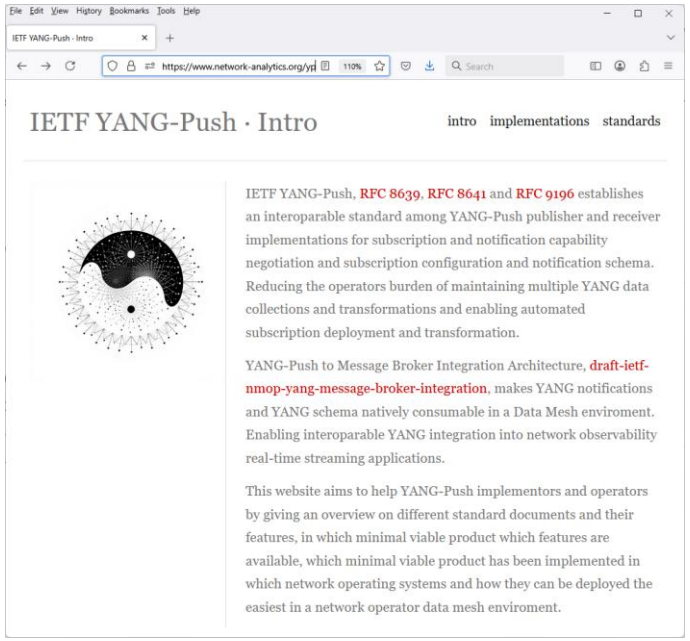
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 5 YANG-Push Publishers
- 2 YANG-Push Receivers
- 2 YANG-Push Network Telemetry Message
- 1 YANG Message Broker Producer and Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

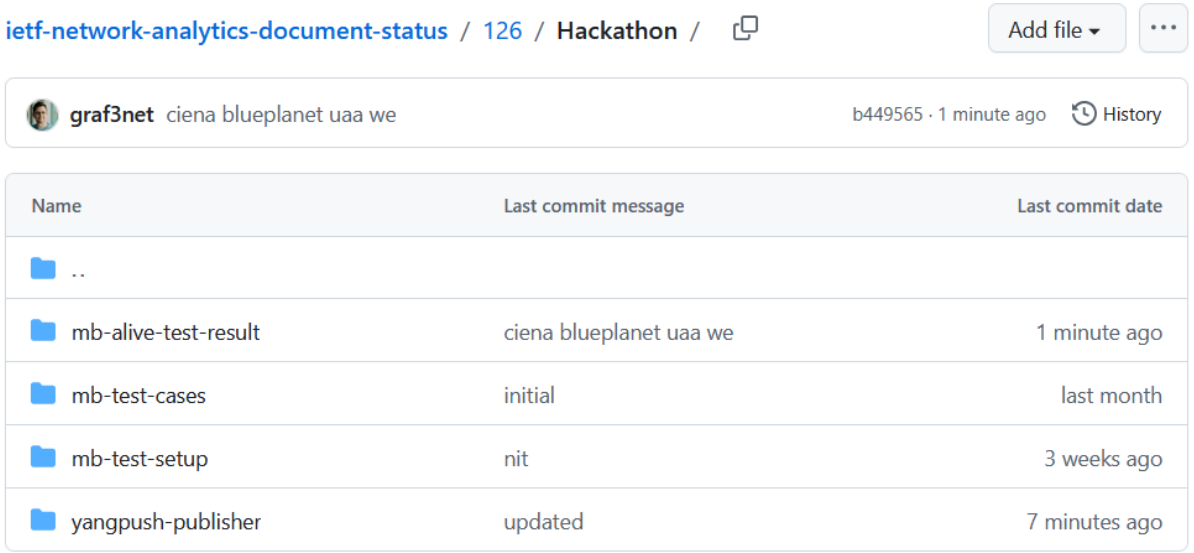
https://www.network-analytics.org/yp/how-to-deploy.html

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router),MA (OLT), Arrcus Arcos and Ductus DMAP
- YANG-Push Receiver – NetGauze™ and Pmacct
- YANG Message Broker Producer – NetGauze™
- YANG Message Broker Consumer – NetGauze™, Ciena Blue Planet UAA Workflow Engine, Cisco Crosswork Ingestion Service
- YANG Schema Registry – Apache Kafka YANG Plugin
- Libyang - https://github.com/CESNET/libyang/releases/tag/v5.4.9
- Yangkit - https://github.com/yang-central/yangkit/
- Wireshark - udp-notif dissector

Hackathon – Repositories

The results of the end-to-end tests

- Netconf RPCs and YANG-Push JSON and CBOR encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation

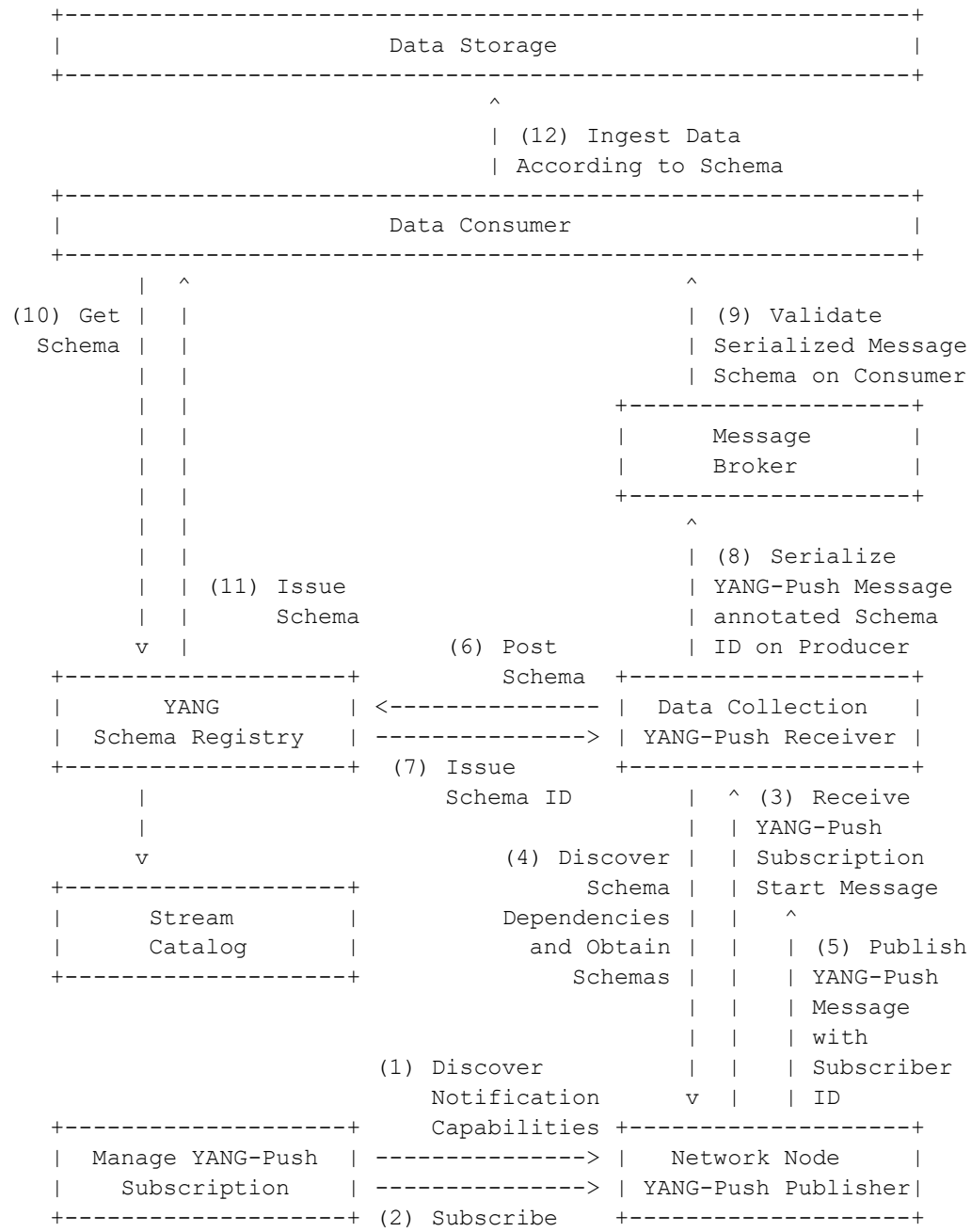


https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/126/Hackathon/

An Architecture for YANG-Push to Message Broker Integration

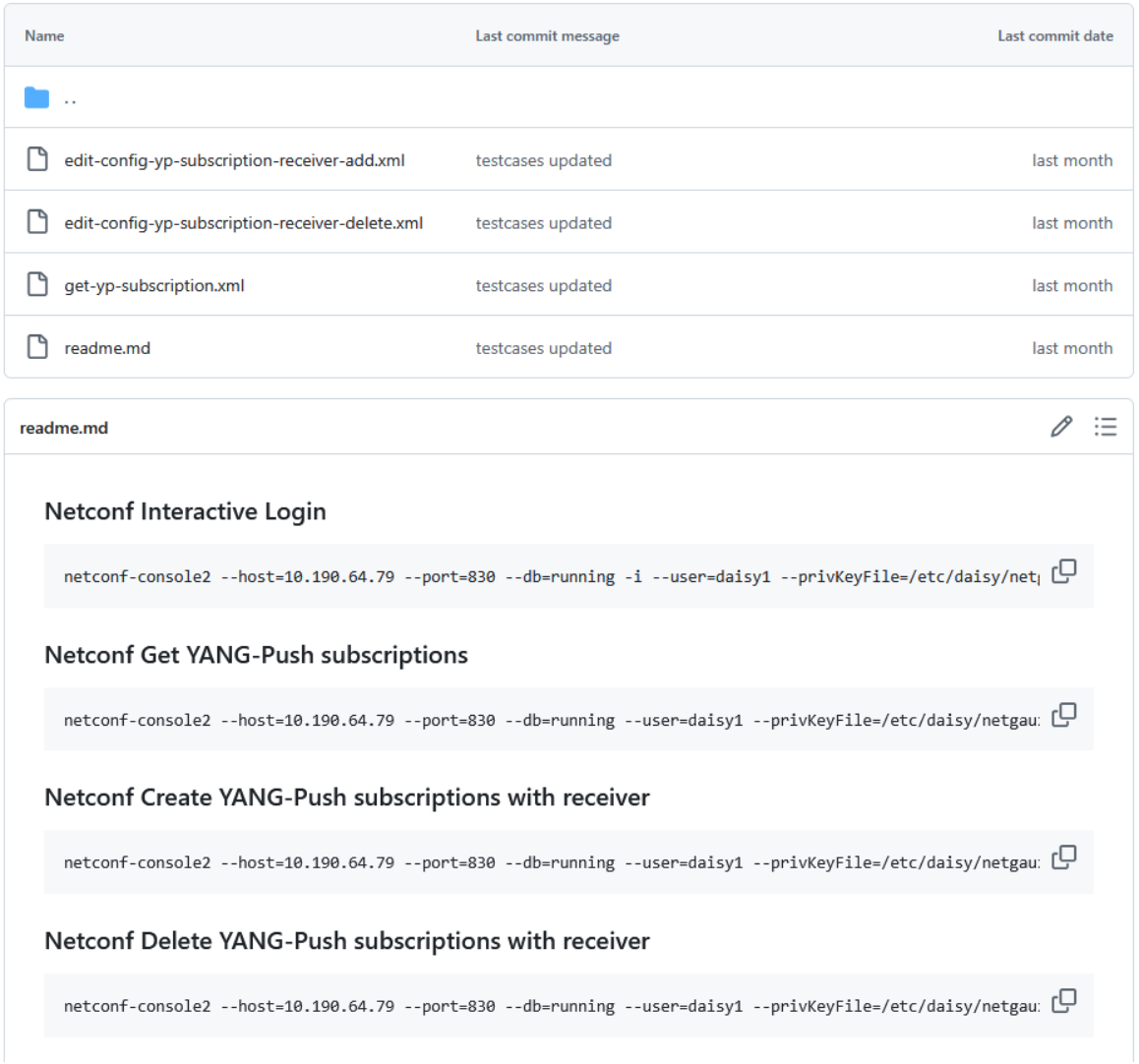
draft-ietf-nmop-yang-message-broker-integration
draft-ietf-nmop-message-broker-telemetry-message
draft-ietf-nmop-yang-message-broker-message-key
draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
  RFC 8639
- Subscription to YANG Notifications for Datastore Updates
  RFC 8641
- UDP-based Transport for Configured Subscriptions
  draft-ietf-netconf-udp-notif
- Subscription to Distributed Notifications
  draft-ietf-netconf-distributed-notif
- Extensible YANG Model for YANG-Push Notifications
  draft-ietf-netconf-notif-envelope
- Support of Versioning in YANG Notifications Subscription
  draft-ietf-netconf-yang-notifications-versioning
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
  RFC 9196
- YANG Notification Transport Capabilities
  draft-ietf-netconf-yp-transport-capabilities
- YANG Library
  RFC 8525
- Augmented-by Addition into the IETF-YANG-Library
  draft-ietf-netconf-yang-library-augmentation
- Validating anydata in YANG Library context
  draft-netana-nmop-yang-anydata-validation
- Encoding of Data Modeled with YANG in the CBOR
  RFC 9254



YANG-Push Publisher - Subscription

- Netconf RPC calls to create, delete and obtain YANG-Push subscriptions and receiver configurations.



Name	Last commit message	Last commit date
..		
edit-config-yp-subscription-receiver-add.xml	testcases updated	last month
edit-config-yp-subscription-receiver-delete.xml	testcases updated	last month
get-yp-subscription.xml	testcases updated	last month
readme.md	testcases updated	last month

readme.md

Netconf Interactive Login

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running -i --user=daisy1 --privKeyFile=/etc/daisy/net
```

Netconf Get YANG-Push subscriptions

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Create YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

Netconf Delete YANG-Push subscriptions with receiver

```
netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau
```

YANG-Push Publisher - Notifications

- Subscription state change notifications as JSON and PCAP.
- Push update and change-update notifications as JSON and PCAP.

Name	Last commit message	Last commit date
..		
1-push-update.json	testcases updated	last month
1-subscription-started.json	testcases updated	last month
1-yp-subscripton.pcap	testcases updated	last month
2-push-update.json	testcases updated	last month
2-subscription-started.json	testcases updated	last month
2-yp-subscripton.pcap	testcases updated	last month
27-push-update.json	testcases updated	last month
27-subscription-started.json	testcases updated	last month
27-yp-subscripton.pcap	testcases updated	last month
readme.md	testcases updated	last month

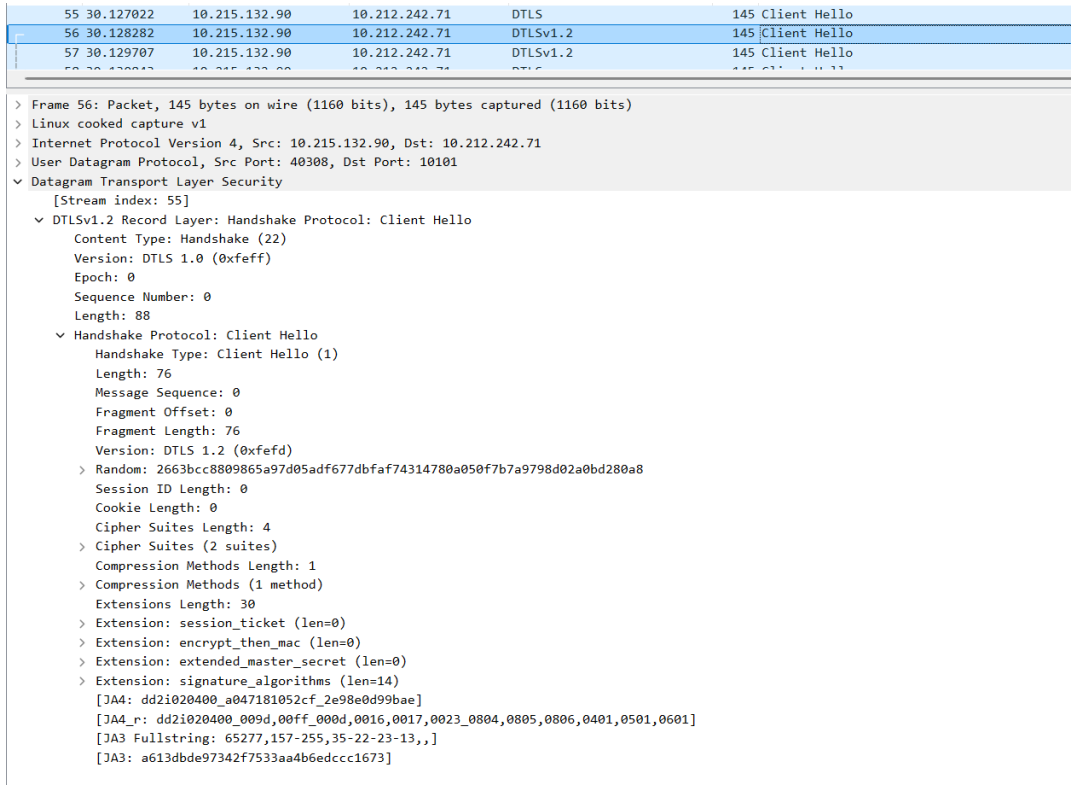
readme.md

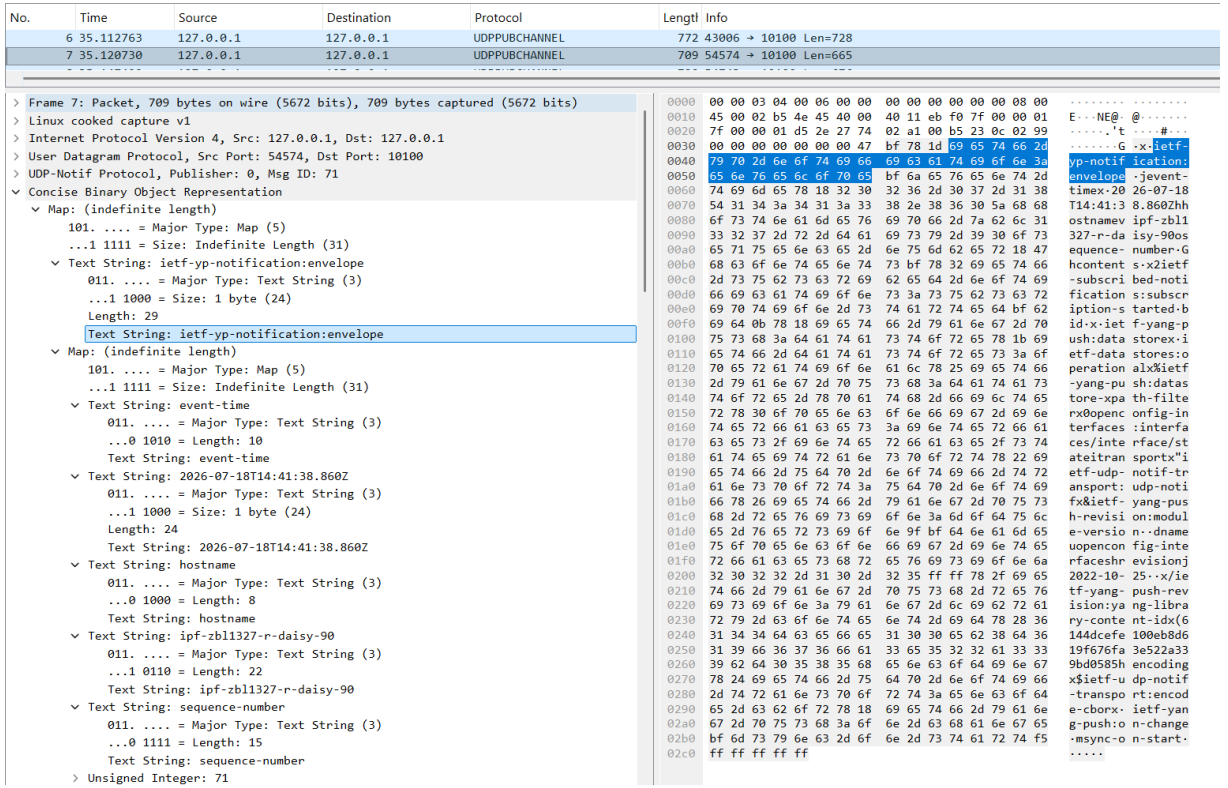
YANG-Push Notifications

subscription-started, push-update and push-change-update notifications in raw and json sorted by subscription id. pcap requires https://github.com/network-analytics/udp-notif-dissector Wireshark dissector.

YANG-Push UDP-Notif – DTLS and CBOR

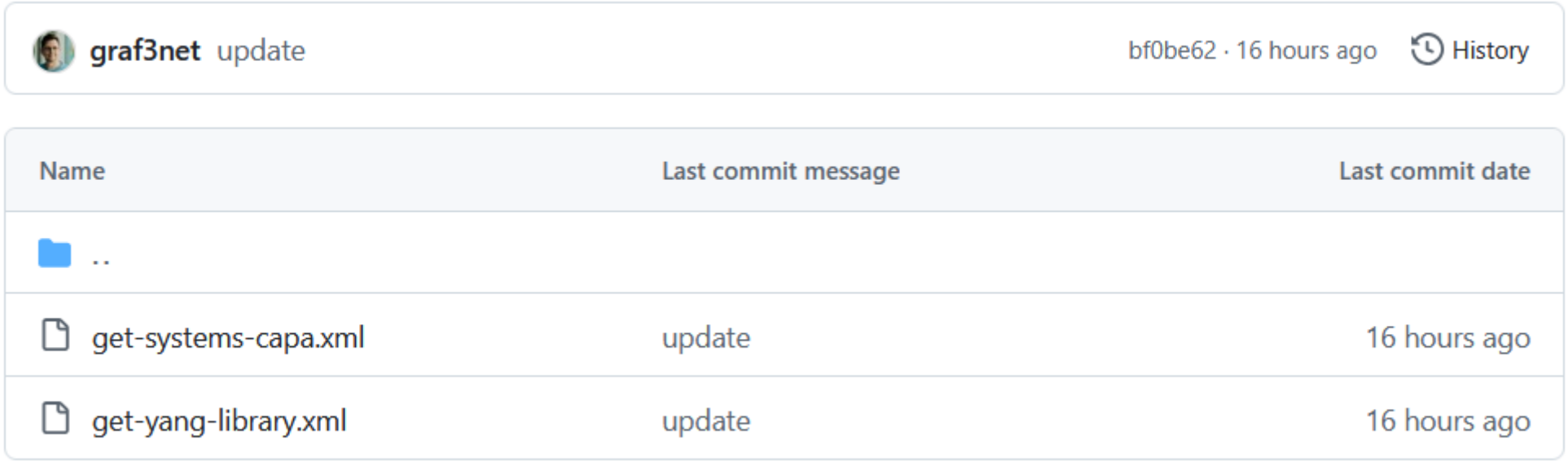
- With Cisco IOS XR, 6Wind VSR and Arrcus Arcos we have now three implementations supporting CBOR Name Identifiers
- With Cisco IOS XR we have the first implementation supporting DTLS 1.2 in UDP-Notif.





YANG-Push Capabilities & YANG Library - RPC

- Netconf XML encoded reply for RFC 9196 YANG-Push capabilities.
- Netconf XML encoded reply for RFC 8525 with draft-ietf-netconf-yang-library-augmentedby.



YANG-Push

IETF 126 Publisher Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open-Source	Ductus DMAP
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓		✓
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif with DTLS				✓			
draft-ietf-netconf-distributed-notif	✓	✓	✓		na		
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓		P
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓	✓		
RFC 8525 YANG Library	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓	✓
RFC 9196 System and Notification Capabilities			✓	P	✓		✓
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓		
RFC 9254 CBOR Named Identifiers	✓			✓	✓		

Green marked describes new capability at IETF 126. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 126 Module Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	✓	✓		
ietf-hardware.yang (inventory, state and statistic)	✓	✓	✓		
ietf-alarms.yang (state)	✓	✓	✓		
ieee802-dot1ab-lldp.yang (inventory and state)	✓	✓			
ieee802-dot1ax.yang (inventory, state and statistic)					
ietf-bfd-ip-sh.yang (state)					
ietf-bfd-ip-mh.yang (state)					
ietf-bfd-lag.yang (state)					
ietf-isis.yang (state)					

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Message Broker

IETF 126 Implementation Status

	NetGauze™	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓			
YANG-Push Receiver	✓	✓			
YANG Schema Retrieval	✓				
YANG Message Broker Producer	✓				
YANG Schema Registry	na	na	✓	na	na
YANG Message Broker Consumer	✓			✓	✓
Validate Telemetry Message against YANG Schema	✓			P	P
draft-ietf-netmod-yang-anydata-validation	✓				
YANG Schema Comparison	na	na	✓	na	na

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Library Feature Comparison

- An extensive YANG feature comparison between the different YANG library open-source implementations.
- Focuses on YANG features needed to validate YANG instance data against YANG schema and performing YANG to YANG transformations.

Table 15. Global synthesis of support levels across evaluated feature families

Feature family	libyang	Yangkit	ODL Yang Tools
Schema loading and validation	Supported	Supported	Partially supported
Module inspection	Partially supported	Supported	Supported
JSON data validation	Supported	Supported	Partially supported
CBOR data validation	Not supported	Supported	Not supported
Anydata validation	Supported	Partially supported	Partially supported
YANG structures	Supported	Supported	Not supported
Anydata inside structures	Supported	Partially supported	Not applicable
Structures inside anydata	Supported	Supported	Not applicable
XPath/data extraction	Supported	Partially supported	Partially supported
Subtree validation	Supported	Supported	Supported
YANG-to-YANG transformation	Supported	Supported for the described add/update cases	Supported for the described add/update cases
YANG Library loading	Supported	Supported	Supported
Yang Push validation	Supported	Supported	Not applicable
Telemetry validation	Supported	Supported	Not applicable
Notification envelope validation	Supported	Supported	Not applicable
Schema comparison	Supported through specific yanglint branch	Implementable	Implementable

https://github.com/network-analytics/yang_tests/blob/master/report_v2.pdf

Next Steps from IETF 125 Hackathon…

- Re-do YANG Message Broker Consumer tests on updated testbed with Ciena Blue Planet UAA Workflow Engine
- Complete IETF YANG module subscription tests with 6Wind VSR
- Compare YANG feature coverage between libyang, yangkit and yangtools
- Revert import only modules for augmented-by workaround
- Preserve cached YANG-Push subscription state across receiver reload
- Obtain YANG-Push subscription state from publisher if missing
- Support Netconf session queuing for schema retrieval
- Support libyang 5.4.9 in YANG-Push Receiver and Message Broker Consumer
- Message Key and Topic Name proof of concept

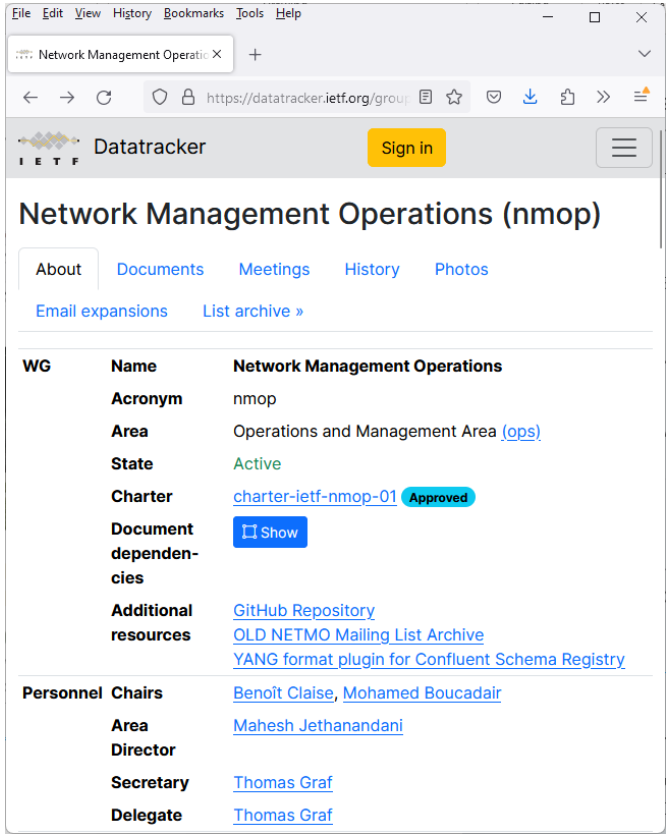


Next Steps for IETF 127 Hackathon…

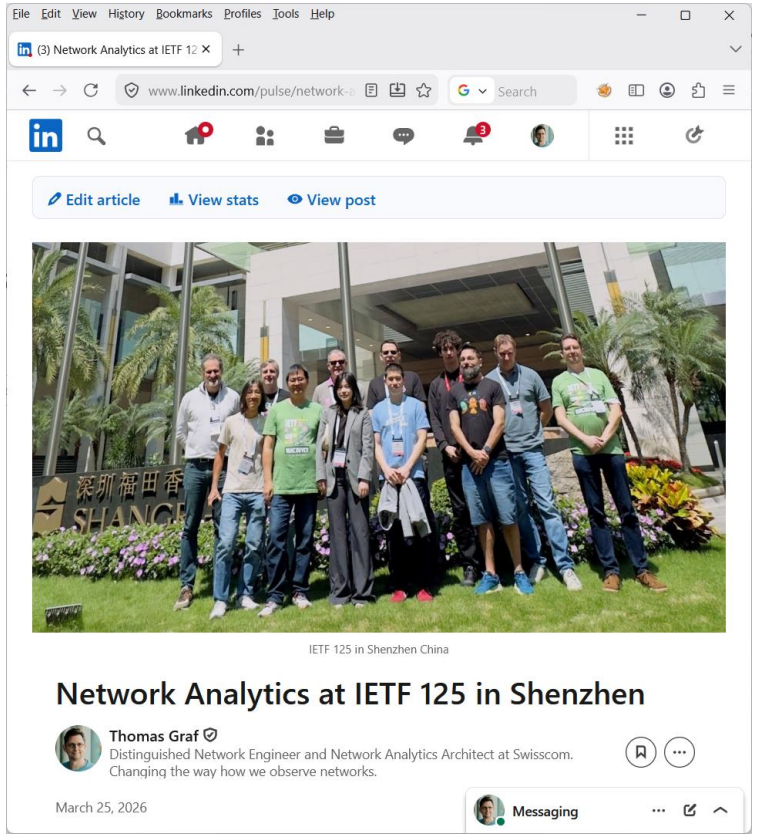
- Support building RFC 8525 YANG Library in Java Message Broker Consumer
- Support draft-netana-nmop-yang-anydata-validation and RFC 8525 YANG library in yangkit
- Support YANG-Push Message Broker Topic Naming with dynamic Apache Kafka topic creation API

IETF NMOP 125/126 – Network Observability Development

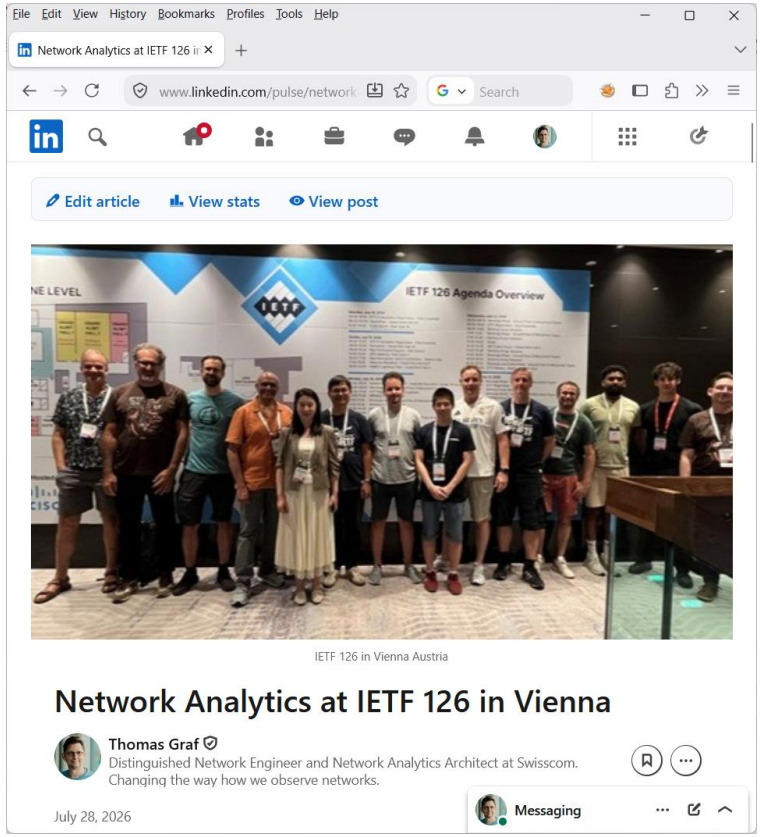
Network Anomaly Detection and YANG-Push/Message Broker Integration



https://datatracker.ietf.org/group/nmop/about/



https://www.linkedin.com/pulse/network-analytics-ietf-125-shenzhen-thomas-graf-svmee/.



https://www.linkedin.com/pulse/network-analytics-ietf-126-vienna-thomas-graf-vng8f/

RFC 8343 – ietf-interfaces.yang vs. openconfig-interfaces.yang

YANG Data Model for Interface Management

```
module: ietf-interfaces
  +--rw interfaces
     +--rw interface* [name]
        +--rw name                            string
        +--rw description?                    string
        +--rw type                            identityref
        +--rw enabled?                        boolean
        +--rw link-up-down-trap-enable?       enumeration {if-mib}?
        +--ro admin-status                    enumeration {if-mib}?
        +--ro oper-status                     enumeration
        +--ro last-change?                    yang:date-and-time
        +--ro if-index                        int32 {if-mib}?
        +--ro phys-address?                   yang:phys-address
        +--ro higher-layer-if*                interface-ref
        +--ro lower-layer-if*                 interface-ref
        +--ro speed?                          yang:gauge64
        +--ro statistics
           +--ro discontinuity-time     yang:date-and-time
           +--ro in-octets?             yang:counter64
           +--ro in-unicast-pkts?       yang:counter64
           +--ro in-broadcast-pkts?     yang:counter64
           +--ro in-multicast-pkts?     yang:counter64
           +--ro in-discards?           yang:counter32
           +--ro in-errors?             yang:counter32
           +--ro in-unknown-protos?     yang:counter32
           +--ro out-octets?            yang:counter64
           +--ro out-unicast-pkts?      yang:counter64
           +--ro out-broadcast-pkts?    yang:counter64
           +--ro out-multicast-pkts?    yang:counter64
           +--ro out-discards?          yang:counter32
           +--ro out-errors?            yang:counter32
```

```
module: openconfig-interfaces
  +--rw interfaces
     +--rw interface* [name]
        +--rw name             -> ../config/name
        +--rw config
        |  +--rw name?            string
        |  +--rw type             identityref
        |  +--rw mtu?             uint16
        |  +--rw loopback-mode?   oc-opt-types:loopback-mode-type
        |  +--rw description?     string
        |  +--rw enabled?         boolean
        +--ro state
        |  +--ro name?            string
        |  +--ro type             identityref
        |  +--ro mtu?             uint16
        |  +--ro loopback-mode?   oc-opt-types:loopback-mode-type
        |  +--ro description?     string
        |  +--ro enabled?         boolean
        |  +--ro ifindex?         uint32
        |  +--ro admin-status     enumeration
        |  +--ro oper-status      enumeration
        |  +--ro last-change?     oc-types:timeticks64
        |  +--ro logical?         boolean
        |  +--ro management?      boolean
        |  +--ro cpu?             boolean
        |  +--ro counters
        |     +--ro in-octets?             oc-yang:counter64
        |     +--ro in-pkts?               oc-yang:counter64
        |     +--ro in-unicast-pkts?       oc-yang:counter64
        |     +--ro in-broadcast-pkts?     oc-yang:counter64
        |     +--ro in-multicast-pkts?     oc-yang:counter64
        |     +--ro in-discards?           oc-yang:counter64
        |     +--ro in-errors?             oc-yang:counter64
        |     +--ro in-unknown-protos?     oc-yang:counter64
        |     +--ro in-fcs-errors?         oc-yang:counter64
        |     +--ro out-octets?            oc-yang:counter64
        |     +--ro out-pkts?              oc-yang:counter64
        |     +--ro out-unicast-pkts?      oc-yang:counter64
        |     +--ro out-broadcast-pkts?    oc-yang:counter64
        |     +--ro out-multicast-pkts?    oc-yang:counter64
        |     +--ro out-discards?          oc-yang:counter64
        |     +--ro out-errors?            oc-yang:counter64
```

```
module: openconfig-interfaces
  +--rw interfaces
     +--rw interface* [name]
     +--rw subinterfaces
        +--rw subinterface* [index]
           +--rw index     -> ../config/index
           +--rw config
           |  +--rw index?         uint32
           |  +--rw description?   string
           |  +--rw enabled?       boolean
           +--ro state
              +--ro index?          uint32
              +--ro description?    string
              +--ro enabled?        boolean
              +--ro name?           string
              +--ro ifindex?        uint32
              +--ro admin-status    enumeration
              +--ro oper-status     enumeration
              +--ro last-change?    oc-types:timeticks64
              +--ro logical?        boolean
              +--ro management?     boolean
              +--ro cpu?            boolean
              +--ro counters
                 +--ro in-octets?             oc-yang:counter64
                 +--ro in-pkts?               oc-yang:counter64
                 +--ro in-unicast-pkts?       oc-yang:counter64
                 +--ro in-broadcast-pkts?     oc-yang:counter64
                 +--ro in-multicast-pkts?     oc-yang:counter64
                 +--ro in-discards?           oc-yang:counter64
                 +--ro in-errors?             oc-yang:counter64
                 +--ro in-unknown-protos?     oc-yang:counter64
                 +--ro in-fcs-errors?         oc-yang:counter64
                 +--ro out-octets?            oc-yang:counter64
                 +--ro out-pkts?              oc-yang:counter64
                 +--ro out-unicast-pkts?      oc-yang:counter64
                 +--ro out-broadcast-pkts?    oc-yang:counter64
                 +--ro out-multicast-pkts?    oc-yang:counter64
                 +--ro out-discards?          oc-yang:counter64
                 +--ro out-errors?            oc-yang:counter64
```

RFC 8348 – ietf-hardware.yang vs. openconfig-platform.yang

YANG Data Model for Hardware Management

```
module: ietf-hardware
  +--rw hardware
     +--ro last-change?   yang:date-and-time
     +--rw component* [name]
        +--rw name                  string
        +--rw class                 identityref
        +--ro physical-index?       int32 {entity-mib}?
        +--ro description?          string
        +--rw parent?               -> ../../component/name
        +--rw parent-rel-pos?       int32
        +--ro contains-child*       -> ../../component/name
        +--ro hardware-rev?         string
        +--ro firmware-rev?         string
        +--ro software-rev?         string
        +--ro serial-num?           string
        +--ro mfg-name?             string
        +--ro model-name?           string
        +--rw alias?                string
        +--rw asset-id?             string
        +--ro is-fru?               boolean
        +--ro mfg-date?             yang:date-and-time
        +--rw uri*                  inet:uri
        +--ro uuid?                 yang:uuid
        +--rw state {hardware-state}?
        |  +--ro state-last-changed?   yang:date-and-time
        |  +--rw admin-state?          admin-state
        |  +--ro oper-state?           oper-state
        |  +--ro usage-state?          usage-state
        |  +--ro alarm-state?          alarm-state
        |  +--ro standby-state?        standby-state
        +--ro sensor-data {hardware-sensor}?
           +--ro value?                    sensor-value
           +--ro value-type?               sensor-value-type
           +--ro value-scale?              sensor-value-scale
           +--ro value-precision?          sensor-value-precision
           +--ro oper-status?              sensor-status
           +--ro units-display?            string
           +--ro value-timestamp?          yang:date-and-time
           +--ro value-update-rate?        uint32
```

```
module: openconfig-platform
  +--rw components
     +--rw component* [name]
        +--rw name                                -> ../config/name
        +--rw config
        |  +--rw name?   string
        +--ro state
        |  +--ro name?                            string
        |  +--ro type?                            union
        |  x--ro id?                              string
        |  x--ro location?                        string
        |  +--ro install-position?                string
        |  +--ro install-component?               -> ../name
        |  +--ro description?                     string
        |  +--ro mfg-name?                        string
        |  +--ro mfg-date?                        oc-yang:date
        |  +--ro hardware-version?                string
        |  +--ro firmware-version?                string
        |  +--ro software-version?                string
        |  +--ro serial-no?                       string
        |  +--ro part-no?                         string
        |  +--ro model-name                       string
        |  +--ro clei-code?                       string
        |  +--ro removable?                       boolean
        |  +--ro oper-status?                     identityref
        |  +--ro empty?                           boolean
        |  +--ro parent?                          -> ../../../component/config/name
        |  +--ro redundant-role?                  oc-platform-types:component-redundant-role
        |  +--ro last-poweroff-reason
        |  |  +--ro trigger?   component-last-poweroff-reason-trigger
        |  |  +--ro details?   string
        |  +--ro last-poweroff-time?              oc-types:timeticks64
        |  +--ro last-switchover-reason
        |  |  +--ro trigger?   component-redundant-role-switchover-reason-trigger
        |  |  +--ro details?   string
        |  +--ro last-switchover-time?            oc-types:timeticks64
        |  +--ro last-reboot-reason?              identityref
        |  x--ro last-reboot-time?                oc-types:timeticks64
        |  +--ro boot-time?                       oc-types:timeticks64
        |  +--ro switchover-ready?                boolean
        |  +--ro base-mac-address?                oc-yang:mac-address
        |  +--ro temperature
        |  |  +--ro instant?                      decimal64
        |  |  +--ro avg?                          decimal64
        |  |  +--ro min?                          decimal64
        |  |  +--ro max?                          decimal64
        |  |  +--ro alarm-status?                 boolean
        |  |  +--ro alarm-threshold?   uint32
        |  |  +--ro alarm-severity?    identityref
        |  +--ro memory
        |  |  +--ro available?   uint64
        |  |  +--ro utilized?    uint64
        |  +--ro allocated-power?                 uint32
        |  +--ro used-power?                      Uint32
        +--rw chassis
        |  +--rw config
        |  +--ro state
        |  +--rw utilization
        |     +--rw resources
        |        +--rw resource* [name]
        |           +--rw name      -> ../config/name
        |           +--rw config
        |           |  +--rw name?                          string
        |           |  +--rw used-threshold-upper?          oc-types:percentage
        |           |  +--rw used-threshold-upper-clear?    oc-types:percentage
        |           +--ro state
        |              +--ro name?                          string
        |              +--ro used-threshold-upper?          oc-types:percentage
        |              +--ro used-threshold-upper-clear?    oc-types:percentage
        |              +--ro used?                          uint64
        |              +--ro committed?                     uint64
        |              +--ro free?                          uint64
        |              +--ro max-limit?                     uint64
        |              +--ro high-watermark?                uint64
        |              +--ro last-high-watermark?           oc-types:timeticks64
        |              +--ro used-threshold-upper-exceeded? boolean
        +--rw power-supply
        |  +--rw config
        |  +--ro state
        +--rw fan
        |  +--rw config
        |  +--ro state
        |     +--ro oc-fan:speed?   uint32
        +--rw fabric
        |  +--rw config
        |  +--ro state
        +--rw storage
        |  +--rw config
        |  +--ro state
        +--rw cpu
        |  +--rw config
        |  +--ro state
        |  +--rw oc-cpu:utilization
        |     +--ro oc-cpu:state
        +--rw integrated-circuit
        |  +--rw config
        |  +--ro state
        |  +--rw utilization
        |     +--rw resources
        |        +--rw resource* [name]
        |           +--rw name      -> ../config/name
        |           +--rw config
        |           |  +--rw name?                          string
        |           |  +--rw used-threshold-upper?          oc-types:percentage
        |           |  +--rw used-threshold-upper-clear?    oc-types:percentage
        |           +--ro state
        |              +--ro name?                          string
        |              +--ro used-threshold-upper?          oc-types:percentage
        |              +--ro used-threshold-upper-clear?    oc-types:percentage
        |              +--ro used?                          uint64
        |              +--ro committed?                     uint64
        |              +--ro free?                          uint64
        |              +--ro max-limit?                     uint64
        |              +--ro high-watermark?                uint64
        |              +--ro last-high-watermark?           oc-types:timeticks64
        |              +--ro used-threshold-upper-exceeded? boolean
        +--rw oc-transceiver:transceiver
        |  +--rw oc-transceiver:config
        |  |  +--rw oc-transceiver:enabled?                 boolean
        |  |  +--rw oc-transceiver:form-factor-preconf?     identityref
        |  |  +--rw oc-transceiver:ethernet-pmd-preconf?    identityref
        |  |  +--rw oc-transceiver:fec-mode?                identityref
        |  |  +--rw oc-transceiver:module-functional-type?  identityref
        |  +--ro oc-transceiver:state
        |  |  +--ro oc-transceiver:enabled?                 boolean
        |  |  +--ro oc-transceiver:form-factor-preconf?     identityref
        |  |  +--ro oc-transceiver:ethernet-pmd-preconf?    identityref
        |  |  +--ro oc-transceiver:fec-mode?                identityref
        |  |  +--ro oc-transceiver:module-functional-type?  identityref
        |  |  +--ro oc-transceiver:present?                 enumeration
        |  |  +--ro oc-transceiver:form-factor?             identityref
        |  |  +--ro oc-transceiver:connector-type?          identityref
        |  |  +--ro oc-transceiver:vendor?                  string
        |  |  +--ro oc-transceiver:vendor-part?             string
        |  |  +--ro oc-transceiver:vendor-rev?              string
        |  |  +--ro oc-transceiver:ethernet-pmd?            identityref
        |  |  +--ro oc-transceiver:sonet-sdh-compliance-code?   identityref
        |  |  +--ro oc-transceiver:otn-compliance-code?     identityref
        |  |  +--ro oc-transceiver:serial-no?               string
        |  |  +--ro oc-transceiver:date-code?               oc-yang:date-and-time
        |  |  +--ro oc-transceiver:fault-condition?         boolean
        |  |  +--ro oc-transceiver:fec-status?              identityref
        |  |  +--ro oc-transceiver:fec-uncorrectable-blocks?    yang:counter64
        |  |  +--ro oc-transceiver:fec-uncorrectable-words?     yang:counter64
        |  |  +--ro oc-transceiver:fec-corrected-bytes?         yang:counter64
        |  |  +--ro oc-transceiver:fec-corrected-bits?          yang:counter64
```

RFC 8632/8348 – ietf-alarms/ietf-hardware.yang vs openconfig-alarms.yang

YANG Data Model for Alarm Management

```
  notifications:
    +---n hardware-state-change
    +---n hardware-state-oper-enabled {hardware-state}?
    |  +--ro name?          -> /hardware/component/name
    |  +--ro admin-state?   -> /hardware/component/state/admin-state
    |  +--ro alarm-state?   -> /hardware/component/state/alarm-state
    +---n hardware-state-oper-disabled {hardware-state}?
       +--ro name?          -> /hardware/component/name
       +--ro admin-state?   -> /hardware/component/state/admin-state
       +--ro alarm-state?   -> /hardware/component/state/alarm-state
```

```
 notifications:
    +---n alarm-notification
    |  +--ro resource                resource
    |  +--ro alarm-type-id           alarm-type-id
    |  +--ro alarm-type-qualifier?   alarm-type-qualifier
    |  +--ro alt-resource*           resource
    |  +--ro related-alarm* [resource alarm-type-id alarm-type-
qualifier] {alarm-correlation}?
    |  |  +--ro resource                -> /alarms/alarm-
list/alarm/resource
    |  |  +--ro alarm-type-id           -> /alarms/alarm-
list/alarm[resource=current()/../resource]/alarm-type-id
    |  |  +--ro alarm-type-qualifier    -> /alarms/alarm-
list/alarm[resource=current()/../resource][alarm-type-
id=current()/../alarm-type-id]/alarm-type-qualifier
    |  +--ro impacted-resource*      resource {service-impact-
analysis}?
    |  +--ro root-cause-resource*    resource {root-cause-analysis}?
    |  +--ro time                    yang:date-and-time
    |  +--ro perceived-severity      severity-with-clear
    |  +--ro alarm-text              alarm-text
    +---n alarm-inventory-changed
```

```
module: openconfig-platform
  +--rw components
     +--rw component* [name]
        +--rw name                    -> ../config/name
        +--rw config
        |  +--rw name?   string
        +--ro state
        |  +--ro oc-alarms:equipment-failure?    boolean
        |  +--ro oc-alarms:equipment-mismatch?   boolean
```

IEEE LLDP – ieee802-dot1ab-lldp.yang vs. openconfig-lldp.yang

YANG Data Model for LLDP Neighbor Discovery

```
module: ieee802-dot1ab-lldp
  +--rw lldp
     +--ro local-system-data
     |  +--ro chassis-id-subtype?                 ieee:chassis-id-subtype-type
     |  +--ro chassis-id?                         ieee:chassis-id-type
     |  +--ro system-name?                        string
     |  +--ro system-description?                 string
     |  +--ro system-capabilities-supported?      lldp-types:system-capabilities-map
     |  +--ro system-capabilities-enabled?        lldp-types:system-capabilities-map
     +--rw port* [name dest-mac-address]
        +--rw name                              if:interface-ref
        +--rw dest-mac-address                  ieee:mac-address
        +--ro port-id?                          ieee:port-id-type
        +--ro remote-systems-data* [time-mark remote-index]
           +--ro time-mark                            yang:timeticks
           +--ro remote-index                         uint32
           +--ro remote-too-many-neighbors?           boolean
           +--ro remote-changes?                      boolean
           +--ro chassis-id-subtype?                  ieee:chassis-id-subtype-type
           +--ro chassis-id?                          ieee:chassis-id-type
           +--ro port-id-subtype?                     ieee:port-id-subtype-type
           +--ro port-id?                             ieee:port-id-type
           +--ro port-desc?                           string
           +--ro system-name?                         string
           +--ro system-description?                  string
           +--ro system-capabilities-supported?       lldp-types:system-capabilities-map
           +--ro system-capabilities-enabled?         lldp-types:system-capabilities-map
```

```
module: openconfig-lldp
  +--rw lldp
     +--ro state
     |  +--ro enabled?                       boolean
     |  +--ro hello-timer?                   uint64
     |  +--ro suppress-tlv-advertisement*    identityref
     |  +--ro system-name?                   string
     |  +--ro system-description?            string
     |  +--ro chassis-id?                    string
     |  +--ro chassis-id-type?               oc-lldp-types:chassis-id-type
     +--rw interfaces
        +--rw interface* [name]
           +--rw name          -> ../config/name
           +--ro state
           |  +--ro name?       oc-if:base-interface-ref
           |  +--ro enabled?    boolean
           +--ro neighbors
              +--ro neighbor* [id]
                 +--ro id               -> ../state/id
                 +--ro state
                    |  +--ro system-name?               string
                    |  +--ro system-description?        string
                    |  +--ro chassis-id?                string
                    |  +--ro chassis-id-type?           oc-lldp-types:chassis-id-type
                    |  +--ro id?                        string
                    |  +--ro age?                       uint64
                    |  +--ro last-update?               int64
                    |  +--ro ttl?                       uint16
                    |  +--ro port-id?                   string
                    |  +--ro port-id-type?              oc-lldp-types:port-id-type
                    |  +--ro port-description?          string
                    |  +--ro management-address?        string
                    |  +--ro management-address-type?   string
```

IEEE dot1ax – ieee802-dot1ax.yang vs. openconfig-lacp.yang

YANG Data Model for LAG Interface Management

```
module: ieee802-dot1ax
  +--rw lag-system
     +--rw aggregating-system* [agg-system]
        +--rw agg-system          string
        +--rw system-id?          ieee:mac-address
        +--rw system-priority?    uint32

  augment /if:interfaces/if:interface:
    +--rw aggregator
       +--rw name?                           string
       +--rw agg-system-name?                string
       +--rw admin-state?                    enumeration
       +--rw link-up-down-notification?      enumeration
       +--rw collector-max-delay?            int16
       +--rw aggregator-lacp
          +--rw actor-admin-key?    int16

  augment /if:interfaces/if:interface:
    +--rw aggregation-port
       +--rw aggregation-port-lacp
          +--rw actor-system-priority?           int16
          +--rw actor-admin-key?                 int16
          +--rw partner-admin-system-priority?   int16
          +--rw partner-admin-system-id?         ieee:mac-address
          +--rw partner-admin-key?               int16
          +--rw actor-port-priority?             int16
          +--rw partner-admin-port?              int16
          +--rw partner-admin-port-priority?     int16
          +--rw actor-admin-state?               bits
          +--rw partner-admin-state?             bits
```

```
module: openconfig-lacp
  +--rw lacp
     +--ro state
     |  +--ro system-priority?   uint16
     +--rw interfaces
        +--rw interface* [name]
           +--rw name       -> ../config/name
           +--ro state
           |  +--ro name?              oc-if:base-interface-ref
           |  +--ro interval?          lacp-period-type
           |  +--ro lacp-mode?         lacp-activity-type
           |  +--ro system-id-mac?     oc-yang:mac-address
           |  +--ro system-priority?   uint16
           +--ro members
              +--ro member* [interface]
                 +--ro interface    -> ../state/interface
                 +--ro state
                    +--ro interface?          oc-if:base-interface-ref
                    +--ro activity?           lacp-activity-type
                    +--ro timeout?            lacp-timeout-type
                    +--ro synchronization?    lacp-synchronization-type
                    +--ro aggregatable?       boolean
                    +--ro collecting?         boolean
                    +--ro distributing?       boolean
                    +--ro system-id?          oc-yang:mac-address
                    +--ro oper-key?           uint16
                    +--ro partner-id?         oc-yang:mac-address
                    +--ro partner-key?        uint16
                    +--ro port-num?           uint16
                    +--ro partner-port-num?   uint16
                    +--ro last-change?        oc-types:timeticks64
```

RFC 9127 – ietf-bfd.yang vs. openconfig-bfd.yang

YANG Data Model for Bidirectional Forwarding Detection

BFD IP Single-Hop Hierarchy

https://datatracker.ietf.org/doc/html/rfc9127#section-2.6

```
  notifications:
    +---n singlehop-notification
       +--ro local-discr?                    discriminator
       +--ro remote-discr?                   discriminator
       +--ro new-state?                      state
       +--ro state-change-reason?            iana-bfd-types:diagnostic
       +--ro time-of-last-state-change?      yang:date-and-time
       +--ro dest-addr?                      inet:ip-address
       +--ro source-addr?                    inet:ip-address
       +--ro session-index?                  uint32
       +--ro path-type?                      identityref
       +--ro interface?                      if:interface-ref
       +--ro echo-enabled?                   boolean
```

BFD-over-LAG Hierarchy

https://datatracker.ietf.org/doc/html/rfc9127#section-2

```
  notifications:
    +---n lag-notification
       +--ro local-discr?                    discriminator
       +--ro remote-discr?                   discriminator
       +--ro new-state?                      state
       +--ro state-change-reason?            iana-bfd-types:diagnostic
       +--ro time-of-last-state-change?      yang:date-and-time
       +--ro dest-addr?                      inet:ip-address
       +--ro source-addr?                    inet:ip-address
       +--ro session-index?                  uint32
       +--ro path-type?                      identityref
       +--ro lag-name?                       if:interface-ref
       +--ro member-link?                    if:interface-ref
```

```
module: openconfig-bfd
  +--rw bfd
     +--rw interfaces
        +--rw interface* [id]
           +--rw id                        -> ../config/id
           +--ro state
           |  +--ro id?                               string
           |  +--ro enabled?                          boolean
           |  +--ro local-address?                    oc-inet:ip-address
           |  +--ro desired-minimum-tx-interval?      uint32
           |  +--ro required-minimum-receive?         uint32
           |  +--ro detection-multiplier?             uint8
           |  +--ro enable-per-member-link?           boolean
           +--rw interface-ref
           |  +--ro state
           |     +--ro interface?      -> /oc-if:interfaces/interface/name
           |     +--ro subinterface?   -> /oc-if:interfaces/interface[oc-
if:name=current()/../interface]/subinterfaces/subinterface/index
           +--rw peers
              +--ro peer* [local-discriminator]
                 +--ro local-discriminator    -> ../state/local-discriminator
                 +--ro state
                    +--ro local-address?                      oc-inet:ip-address
                    +--ro remote-address?                     oc-inet:ip-address
                    +--ro subscribed-protocols*               identityref
                    +--ro session-state?                      bfd-session-state
                    +--ro remote-session-state?               bfd-session-state
                    +--ro last-failure-time?                  oc-types:timeticks64
                    +--ro failure-transitions?                uint64
                    +--ro local-discriminator?                string
                    +--ro remote-discriminator?               string
                    +--ro local-diagnostic-code?              bfd-diagnostic-code
                    +--ro remote-diagnostic-code?             bfd-diagnostic-code
                    +--ro remote-minimum-receive-interval?    uint32
                    +--ro demand-mode-requested?              boolean
                    +--ro remote-authentication-enabled?      boolean
                    +--ro remote-control-plane-independent?   boolean
```

RFC 9130 – ietf-isis.yang vs. openconfig-isis.yang

YANG Data Model for the IS-IS Protocol

```
  notifications:
    +---n database-overload
    |  +--ro routing-protocol-name?   -> /rt:routing/
    |  |                                 control-plane-protocols/
    |  |                                 control-plane-protocol/name
    |  +--ro isis-level?              level
    |  +--ro overload?                enumeration
    +---n if-state-change
    |  +--ro routing-protocol-name?   -> /rt:routing/
    |  |                                 control-plane-protocols/
    |  |                                 control-plane-protocol/name
    |  +--ro isis-level?              level
    |  +--ro interface-name?          if:interface-ref
    |  +--ro interface-level?         level
    |  +--ro extended-circuit-id?     extended-circuit-id
    |  +--ro state?                   if-state-type
    +---n adjacency-state-change
    |  +--ro routing-protocol-name?   -> /rt:routing/
    |  |                                 control-plane-protocols/
    |  |                                 control-plane-protocol/name
    |  +--ro isis-level?              level
    |  +--ro interface-name?          if:interface-ref
    |  +--ro interface-level?         level
    |  +--ro extended-circuit-id?     extended-circuit-id
    |  +--ro neighbor?                string
    |  +--ro neighbor-system-id?      system-id
    |  +--ro state?                   adj-state-type
    |  +--ro reason?                  string
```

```
module: openconfig-network-instance
  +--rw network-instances
     +--rw network-instance* [name]
        +--rw name                          -> ../config/name
        +--rw protocols
           +--rw protocol* [identifier name]
              +--rw isis
                 |  +--rw global
                 |  +--rw interfaces
                 |     +--rw interface* [interface-id]
                 |        +--rw interface-id        -> ../config/interface-id
                 |        +--rw levels
                 |           |  +--rw level* [level-number]
                 |           |     +--rw level-number            -> ../config/level-number
                 |           |     +--ro adjacencies
                 |           |     |  +--ro adjacency* [system-id]
                 |           |     |     +--ro system-id    -> ../state/system-id
                 |           |     |     +--ro state
                 |           |     |        +--ro system-id?                      oc-isis-types:system-id
                 |           |     |        +--ro neighbor-ipv4-address?          oc-inet:ipv4-address
                 |           |     |        +--ro neighbor-ipv6-address?          oc-inet:ipv6-address
                 |           |     |        +--ro neighbor-snpa?                  oc-isis-types:snpa
                 |           |     |        +--ro local-extended-circuit-id?      oc-isis-types:extended-circuit-id
                 |           |     |        +--ro neighbor-extended-circuit-id?   oc-isis-types:extended-circuit-id
                 |           |     |        +--ro priority?                       uint8
                 |           |     |        +--ro dis-system-id?                  oc-isis-types:system-id
                 |           |     |        +--ro neighbor-circuit-type?          oc-isis-types:level-type
                 |           |     |        +--ro adjacency-type?                 oc-isis-types:level-type
                 |           |     |        +--ro adjacency-state?                oc-isis-types:isis-interface-adj-state
                 |           |     |        +--ro up-timestamp?                   oc-types:timeticks64
                 |           |     |        +--ro multi-topology?                 boolean
                 |           |     |        +--ro topology*                       identityref
                 |           |     |        +--ro restart-support?                boolean
                 |           |     |        +--ro restart-suppress?               boolean
                 |           |     |        +--ro restart-status?                 boolean
                 |           |     |        +--ro area-address*                   oc-isis-types:area-address
                 |           |     |        +--ro nlpid*                          enumeration
```